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Cardiovascular Aspects of Kidney Disease

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CHAPTER OUTLINE

INTRODUCTION

The expanding availability of dialysis and kidney transplantation in developed countries has led to improvements in the prognosis of patients with end-stage kidney disease (ESKD). However, the life expectancy of patients receiving such renal replacement therapy (RRT) remains considerably lower than that of age- and gender-matched healthy controls, and the risks of both cardiovascular and noncardiovascular causes of mortality are increased at all ages.[1] This chapter will focus on the pathology, epidemiology, and treatment of cardiovascular disease in the chronic kidney disease (CKD) population.

The first description of a link between diseases of the kidneys and the cardiovascular system is attributable to Richard Bright, who described cardiac hypertrophy in patients with small kidneys at postmortem over 170 years ago.[2] The problem became more obvious with the advent of dialysis, because patients survived long enough after ESKD to develop clinical manifestations of cardiovascular disease. In the early 1970s, clinicians were alarmed at the high incidence of cardiovascular events in young patients receiving RRT and presumed that either kidney disease or the dialysis process itself led to accelerated atherosclerosis.[3]

Twenty years later, a task force report published by the U.S. National Kidney Foundation focused the attention of the nephrology community on the problem of increased cardiovascular disease risk associated with CKD.[4] Research over the last 10 years or so has improved our understanding of the cardiovascular consequences of impaired kidney function and has helped suggest appropriate treatment strategies. For example, the nature of the cardiac and arterial changes that complicate CKD are now better described. Although atherosclerosis contributes to arterial pathology in advanced CKD, nonatherosclerotic changes, including arterial wall thickening and calcification, are the dominant pathologic feature.[5] Whereas an increased risk of cardiovascular events associated with atherosclerotic plaque rupture is recognized in patients with CKD,[6] the clinical consequences of structural heart disease (heart failure and sudden dysrhythmic death) appear to be a more important cause of morbidity and mortality than the complications of atherosclerotic disease.[7] However, our understanding remains incomplete.

Although the problem of premature cardiovascular disease was first recognized in dialysis populations, patients with lesser degrees of impaired kidney function are also at increased risk of cardiovascular events.[8] Numerous studies have shown an inverse association between estimated glomerular filtration rate (eGFR) and cardiovascular risk.[9] The nature of both cardiac and arterial disease may change with declining kidney function, and this might have implications for the development of optimal management strategies. It has also become clear that albuminuria is associated with an increased risk of cardiovascular disease, whether or not the GFR is reduced.[9,10]

It follows that the assessment and management of cardiovascular diseases should start early in the course of CKD in an effort to reduce morbidity and mortality, bearing in mind that most patients in whom CKD is diagnosed early do not progress to ESKD. Those who do may have well-established structural cardiac and vascular damage by the time they commence RRT.[11,12] This is particularly important when considering a patient's suitability for transplantation, and most clinicians will screen such patients (or at least those they consider to be at high risk) for cardiovascular disease before transplant listing in an effort to reduce perioperative morbidity and mortality and optimize the results of organ allocation.[13] Despite such screening strategies, patients who receive a kidney transplant remain at higher risk than age- and gender-matched controls without kidney disease.[4]

At present, there is much uncertainty about how to reduce cardiovascular risk in patients with CKD. Epidemiologic relationships among recognized cardiovascular risk factors and particular clinical outcomes are often confounded by other factors associated with ill health in CKD, such as inflammation and malnutrition,[14,15] and it may not be possible to account for them accurately or completely. Consequently, it may be more appropriate to approach the problem of preventing cardiovascular disease in CKD by defining the types of pathology observed in patients with CKD and then considering what is known about the treatment or prevention of such pathologies from studies among individuals without CKD. These provide more reliable information on the effectiveness of particular drugs in specific disease processes. Another possible strategy, which is applicable when treatments that modify a particular risk factor are available, is to conduct randomized trials of such treatments, which can yield unbiased assessments of the causal relevance of that risk factor for a particular type of cardiovascular disease.

Box 54.1 Characteristics of Cardiovascular Diseases in Chronic Kidney Disease

Arteries

Increased wall thickness
Arterial stiffness
Endothelial dysfunction
Arterial calcification

Heart

Altered cardiac geometry
Myocardial fibrosis
Left ventricular dysfunction
Valvular disease
Dysrhythmia and conduction defects

SPECTRUM OF CARDIOVASCULAR PATHOLOGY IN CHRONIC KIDNEY DISEASE

Cardiovascular changes observed in patients with CKD can be divided broadly into those that involve the blood vessels (specifically the arteries) and those that involve the heart (Box 54.1). Although a wide spectrum of cardiovascular pathologies is recognized in CKD, much less is known about the association between particular pathologies and specific cardiovascular risk factors in the CKD population. It is likely that any associations between a given risk factor and particular cardiovascular pathologies will vary in their strength (and possibly direction), so careful phenotyping of cardiovascular outcomes is essential in epidemiologic studies.

ARTERIAL DISEASE

The term *arteriosclerosis* (derived from the Greek, meaning hardening of the arteries) is generally used to describe a range of pathologic processes. Strictly speaking, arteriosclerosis encompasses three different lesions—atherosclerosis, arteriolosclerosis, and Mönckeberg medial calcific sclerosis (or Mönckeberg sclerosis).[16] Atherosclerosis, a word derived from the Greek "atheroma" (gruel-like material), is characterized by the development of lipid-enriched plaques in the intimal layer of the artery. Calcification is an important feature of atherosclerosis, and its presence or absence is relevant in determining the stage of the lesion.[17] Distinct from atherosclerosis, the phenomenon of noncalcified, nonatheromatous stiffening of smaller muscular arteries was first described in patients with Bright disease in 1868 and soon came to be known as "arteriolosclerosis." Then, 35 years later, in 1903, Mönckeberg reported what he considered to be a third distinct form of arterial disease involving the media of the artery, characterized by medial thickening and heavy calcification, without the presence of atheroma.[18]

Patients with CKD may exhibit all the previous features of arteriosclerosis. The exact nature of arterial disease in any given individual patient is likely to depend on multiple factors, including age, exposure to risk factors, and cause and duration of CKD. For example, a patient with preexisting atherosclerosis who develops CKD as a result of atherosclerotic renovascular disease may have a rather different spectrum of arterial pathology than a patient with a similar level of kidney function caused by kidney damage from glomerulonephritis. The various structural and functional arterial abnormalities associated with CKD are covered in more detail later.

ARTERIAL WALL THICKENING

One of the few autopsy studies examining arterial pathology in ESKD has reported that there is more pronounced medial thickening in sections of coronary artery than in equivalent sections from age- and gender-matched, non-CKD patients known to have had coronary artery disease (CAD).[5] Thickening of the arterial wall can be measured noninvasively by assessing the combined width of the intima and media of the carotid artery on ultrasound (carotid intima-media thickening [CIMT]) and has been found to be increased in populations at high risk of cardiovascular disease, such as older adults[19] and patients with type 2 diabetes mellitus.[20] Studies in patients on hemodialysis dating back to the mid-1990s have indicated higher arterial intima-media thickness values for carotid and femoral sites as compared with healthy controls.[21] More recent studies have suggested that increased CIMT is also found in patients with less advanced CKD, indicating that eGFR independently predicts carotid diameter, which increases wall stress.[22] Carotid IMT has also been shown to be a strong predictor of death from cardiovascular causes in patients with CKD, independent of other risk factors.[23,24]

ARTERIAL STIFFENING

Stiffening is thought to be a functional consequence of artery wall thickening (and calcification) and is readily assessed noninvasively by measuring the velocity of propagation of a pulse wave through the arterial tree (pulse wave velocity [PWV]).[25] Stiffening may represent an early feature of CKD-associated arterial disease and can be detected in pediatric populations receiving dialysis as early as the first decade of life.[26] Stiffness has prognostic significance; both carotid[27] and aortic[28] stiffness independently predict death in adult patients on hemodialysis, as in other high-risk groups, such as patients with diabetes mellitus[29] and older adults.[30]

Another method of measuring arterial stiffness is to assess the pulse waveform in an accessible artery (e.g., the carotid artery). A parameter derived from the pulse waveform, the augmentation index, provides a measure of the interaction between outgoing and reflected pulse waveforms at the point of measurement and, in part, reflects the stiffness of the arterial tree.[31] The augmentation index assessed at the common carotid artery has been found to predict mortality in one study of ESKD,[32] but not in another.[33] A further study comparing different methods of measuring arterial stiffness has found that a higher PWV, but not the augmentation index, predicts an increased risk of cardiovascular disease outcomes in patients with CKD.[34]

A third method, the measurement of carotid artery stiffness parameter β, which is determined by monitoring pulsatile changes in the artery during echo-tracking sonography,[35] was used in a prospective cohort study examining the independent predictive value of arterial stiffness in 423 patients on hemodialysis. This measurement of stiffness independently predicted cardiovascular events, even after adjustments had been made for arterial thickness, possibly implicating distinct roles for stiffening and thickening in the development of arterial complications in patients with CKD.[36]

ENDOTHELIAL DYSFUNCTION

The vascular endothelium plays a key role in maintaining arterial tone, predominantly through the continuous production of nitric oxide. Nitric oxide is a vasoactive compound that contributes to the resting tone of the artery and protects against the development of arterial disease by inhibiting vascular smooth muscle cell proliferation, platelet aggregation, and monocyte adhesion.[37] The production of nitric oxide is stimulated by hypoxia, increased sheer stress, or by locally released mediators such as acetylcholine.

Endothelial function can be measured by assessing the vasodilatory response of an artery to endothelial stimulation. This can be achieved by directly infusing compounds into an artery (usually the brachial) or by monitoring the response to reactive hyperemia following temporary arterial occlusion. Arterial vasodilation can be assessed by measuring changes in forearm size using strain gauge plethysmography, which works on the principle that the rate of distention of a forearm is proportional to the rate of arterial inflow, or by measuring the diameter of the arterial lumen using high-resolution ultrasonography.[38]

Studies in patients with stage 4 or 5 CKD have consistently demonstrated impairment of endothelial function by both invasive[39] and noninvasive approaches.[40] One possible mechanism for endothelial dysfunction in the context of CKD is the accumulation of asymmetric dimethylarginine (ADMA), an endogenous inhibitor of the enzyme nitric oxide synthetase.[41] ADMA inhibits nitric oxide synthetase and thereby limits the bioavailability of nitric oxide, which is essential for normal endothelial function.[42] Blood concentrations of ADMA are elevated in CKD, are inversely proportional to GFR,[41,43] and appear to be associated with an increased risk of cardiovascular disease in the general population in some but not all studies.[44] At present, there is no intervention that can selectively reduce ADMA concentrations, so the causal relationship (and clinical relevance) of this risk factor remain(s) unclear.

ARTERIAL CALCIFICATION

Arterial calcification is a recognized feature of two of the pathologic processes that are prevalent in patients with CKD—namely, atherosclerosis[17] and Mönckeberg sclerosis (see earlier).[18] The pattern observed in atherosclerosis is of patchy intimal calcification in association with lipid deposits, whereas in Mönckeberg disease there is a linear pattern of medial calcium deposition (Fig. 54.1).[45] Both are closely linked to disturbances in calcium and phosphate homeostasis and to associated abnormalities in bone mineral metabolism (see Chapter 53).

Autopsy studies have indicated a greater degree of arterial calcification in patients with ESRD than controls with known CAD.[5] In life, calcification can be detected by both ultrasound- and x-ray–based techniques.[46] Studies using electron beam computed tomography, a technique that allows rapid image acquisition, thereby “freezing” the heart during diastole,[47] have indicated that, compared with patients known to have cardiovascular disease, patients on dialysis have much higher calcification scores.[48] Furthermore, calcification develops at a younger age in CKD than in non-CKD populations,[49] being detectable even in children and adolescents.[50]

Some investigators have attempted to distinguish the two patterns of arterial calcification on the basis of imaging studies.[51] Although they may represent a continuum of the same pathologic process,[52] patchy calcium deposition (suggestive of an atherosclerotic pattern) was more common in older patients with a clinical history of cardiovascular events prior to starting dialysis.[51] The physiologic consequences of arterial calcification may include stiffening of the artery,[53] but it is unclear whether calcification has a positive or negative impact on plaque stability.[54]

CARDIAC DISEASE

Echocardiographic studies performed in the mid-1990s indicated a high prevalence of structural heart abnormalities in patients starting dialysis, with 74% having increased left ventricular mass in one study.[11] Left ventricular remodeling occurs well before the initiation of dialysis[55] and is detectable even in patients with stage 2 or 3 CKD.[56,57] Although adaptive in the early stages, such structural changes may eventually lead to functional impairment, including reduced compliance of the left ventricular wall during diastole (diastolic dysfunction), impaired myocardial contractility (systolic dysfunction), or both.[58] In addition to these changes in left ventricular geometry, histologic changes such as fibrosis and calcification occur in the myocardium,[59] and valvular calcification is also frequently observed.[60]

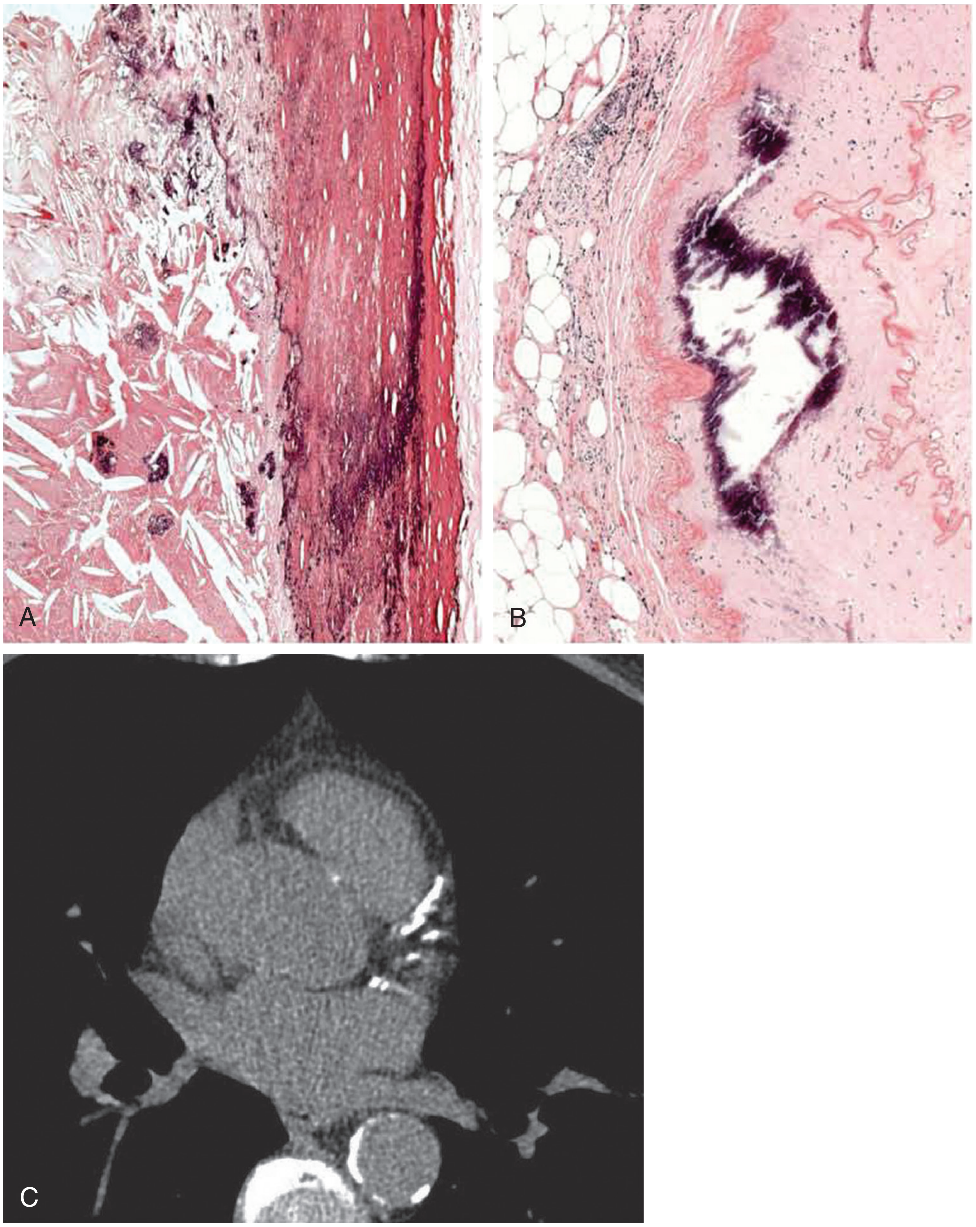

Fig. 54.1 Arterial calcification in chronic kidney disease (CKD). Shown are cross-sections of medium-sized arteries from a patient with CKD showing deposition of calcium *(black)* in the intima (A) and media (B) in association with atherosclerosis (von Kossa stain). Calcium deposits may be visible on CT scanning of the heart as depicted in (C), where calcification is visible in the left anterior descending and left circumflex coronary artery, as well as the descending aorta. (Courtesy AJ Howie.)

ALTERED CARDIAC GEOMETRY

Left ventricular hypertrophy is generally classified according to the predominant pattern of abnormality on echocardiogram. In one study of 3487 patients with CKD, the prevalence of left ventricular hypertrophy was 32%, 48%, 57%, and 75% for eGFR categories 60 or higher, 45–59, 30–44, and less than 30 mL/min/1.73 m^2 (Fig. 54.2).[56] After adjustment for numerous potential confounders, the odds of having left ventricular hypertrophy were more than double among patients with an eGFR less than 30 mL/min/1.73 m^2 compared with those with an eGFR 60 or higher mL/min/1.73 m^2 (odds ratio, 2.2; 95% confidence interval [CI] 1.4 to 3.4). Left ventricular hypertrophy is also recognized in children in whom other confounding diseases are less common (see Fig. 54.2).[61]

Recognizing that categorization is complicated by volume changes related to dialysis, Foley and associates found that 44% of patients had predominantly left ventricular wall thickening (concentric hypertrophy), and 30% had predominantly increased cavity volume (eccentric hypertrophy) in a study of patients starting dialysis.[11] Such changes are likely to represent adaptations to volume and pressure overload.[62] Volume overload increases left ventricular filling pressure and thereby stretches the ventricular wall. The heart adapts by lengthening existing myocytes, thus enlarging the internal dimensions of the left ventricular cavity. This process is usually accompanied by wall thickening, a further adaptive response that reduces wall stress. Thus volume overload results in a ventricle with a thickened wall and enlarged cavity, but with a normal wall thickness–to–internal diameter ratio (eccentric hypertrophy). In contrast, pressure overload increases wall stress during systole, leading to myocyte proliferation and wall thickening, with preservation or reduction of cavity volume (concentric hypertrophy; Fig. 54.3). The previous adaptive responses, which may be reversible in the early stages, are essentially beneficial, at least initially. Dilation permits increased cardiac output for a similar level of energy expenditure, whereas wall thickening redistributes increased tension over a larger area and reduces energy consumption/myocyte.[63] Nevertheless, despite such adaptation, volume overload has recently been shown to be associated with poor outcomes (independently of any effect on blood pressure) in a large cohort of hemodialysis patients.[64]

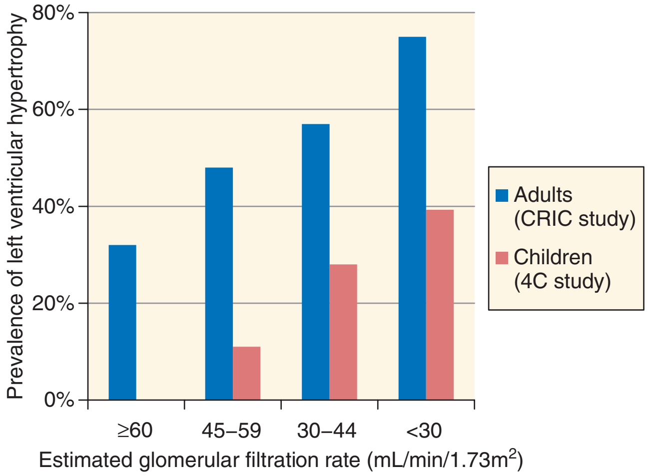

Fig. 54.2 **Prevalence of left ventricular hypertrophy in adults and children by estimated glomerular filtration rate.** (From Park M, Hsu CY, Li Y, et al. Associations between kidney function and subclinical cardiac abnormalities in CKD. *J Am Soc Nephrol* 2012;23:1725-1734; Schaefer F, Doyon A, Azukaitis K, et al. Cardiovascular phenotypes in children with CKD: The 4C Study. *Clin J Am Soc Nephrol* 2017;12:19-28.)

Cardiac magnetic resonance studies have allowed the geometry of the heart to be assessed in a volume-independent manner. Such studies have identified two major types of cardiomyopathy in patients with advanced CKD.[65] Gadolinium-based contrast studies have shown that over two-thirds of patients have left ventricular hypertrophy, with preserved systolic volume and function, which is associated with diffuse myocardial fibrosis. Left ventricular dilation and impaired systolic function were observed in another 15% of patients but, by contrast, this was associated strongly with traditional atherosclerotic risk factors and a higher burden of CAD on angiography.[65] However, the recognition of gadolinium-induced nephrogenic systemic fibrosis has meant that such studies can no longer be conducted until a safer contrast agent has been identified.

IMPAIRED ANGIOGENESIS

Maintaining perfusion to a hypertrophied left ventricle requires capillary angiogenesis, but this is disturbed in CKD—hence the term "impaired angio-adaptation." In the rat model of subtotal nephrectomy, capillary rarefaction begins

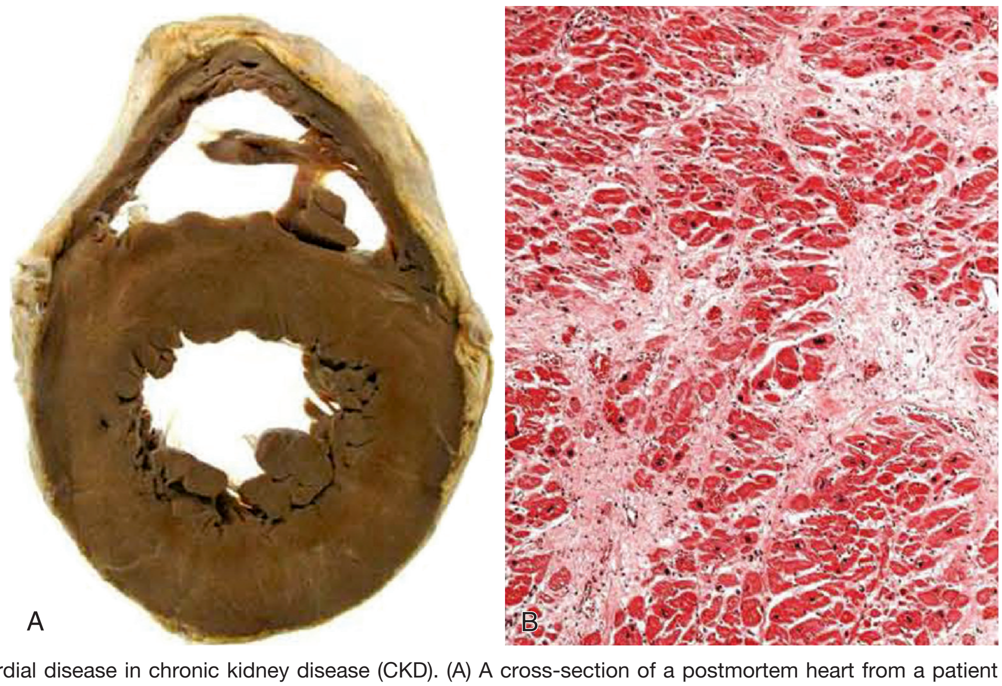

Fig. 54.3 Myocardial disease in chronic kidney disease (CKD). (A) A cross-section of a postmortem heart from a patient with long-standing CKD showing concentric left ventricular hypertrophy. (B) Histologic analysis often reveals myocardial fibrosis (pale staining) disrupting the normal architecture of cardiac myocytes. (A, Courtesy AJ Howie; B, Courtesy Dr. M. Rubens.)

at about 8 to 12 weeks after induction and is progressive.[66] Consequently, such myocardium is more susceptible to an ischemic insult, as demonstrated by the increased volume of myocardium that infarcts in this rat model compared with a healthy rat with a similar ischemic insult.[66] Patients with CKD demonstrate more ischemia on stress than patients with preserved kidney function, and this difference is not simply explained by the burden of traditional cardiovascular risk factors or of coronary atheroma.[67] The cause of the reduced angiogenesis is unclear, although the fact that other tissues such as skeletal muscle and the skin are similarly affected suggests an imbalance in systemic angiogenic factors, such as vascular endothelial growth factor (VEGF) and its receptors, in particular soluble fms-like tyrosine kinase-1 (sFlt-1).[68]

MYOCARDIAL FIBROSIS

In the longer term, excess myocyte work leads to cell death and interstitial cardiac fibrosis.[59] Such maladaptive changes may be exacerbated by ischemia. Even in the absence of occlusive coronary artery lesions, there may be a reduction in capillary density to hypertrophied cardiac myocytes, which exacerbates local hypoxia.[69] Furthermore, stiffening of conduit arteries leads to a fall in diastolic pressure, which in turn may compromise coronary artery perfusion during diastole. Finally, it is possible that repetitive myocardial ischemia induced by hemodialysis exacerbates myocyte injury.[70] A mismatch in oxygen supply and demand to the myocardium might explain the well-recognized clinical observation that patients on dialysis are prone to develop angina, even in the absence of occlusive lesions in the major epicardial coronary arteries (i.e., demand ischemia).[71]

An alternative explanation for the changes described is that patients develop a distinct uremic cardiomyopathy, defined as a primary disease of cardiac muscle associated with CKD and causing systolic dysfunction.[72] Characteristic histologic changes include interstitial myocardial fibrosis. Such pathologic changes may occur early in the course of CKD[73] and could be the result of metabolic changes (e.g., elevations in fibroblast growth factor 23 [FGF23] concentrations; see later, "Fibroblast Growth Factor 23") or neurohormonal adaptations (e.g., activation of the renin-angiotensin system), rather than a response to changes in left ventricular geometry.[74]

CHANGES TO LEFT VENTRICULAR FUNCTION

In the early stages of left ventricular remodeling, indices of systolic function may be preserved by the Starling mechanism and increased sympathetic activity.[59] However, in the longer term, impaired myocardial contractility ensues, together with a reduction in ejection fraction (or systolic dysfunction). Echocardiographic studies have suggested that systolic dysfunction is present in approximately 20% of patients on maintenance dialysis.[75] The disease is clinically silent in the early stages, but in due course patients may develop symptoms suggestive of congestive cardiac failure. Furthermore, symptoms may result from diastolic dysfunction (see later) in patients with a well-preserved ejection fraction.[76] Severe systolic dysfunction may lead to a fall in systolic blood pressure in a previously hypertensive patient, a phenomenon that may help explain the association between lower systolic pressure and mortality in patients on dialysis (see "Blood Pressure").[77] Observational studies have suggested that systolic dysfunction improves following successful kidney transplantation, even in patients with severe disease.[78] Thus the practice of excluding such individuals from transplant waiting lists should be questioned, particularly if prolonged dialysis or CKD exacerbates myocardial dysfunction.[79]

The histologic changes to the myocardium described previously undoubtedly contribute to diastolic dysfunction, which is characterized by impaired ventricular relaxation and reduced ventricular compliance. Compared with systolic dysfunction, diastolic dysfunction is more likely to lead to clinical manifestations such as heart failure. A patient with a stiff left ventricle may be particularly sensitive to tachyarrhythmia (e.g., atrial fibrillation) or an increase in intravascular volume and pulmonary edema, whereas intravascular volume depletion results in reduced ventricular filling (with a risk of syncope) and hemodynamic instability on dialysis.[80] Diastolic dysfunction is more common than systolic dysfunction in patients on dialysis, being present in approximately 50% of prevalent patients,[81] and may be associated with an even worse prognosis than systolic dysfunction.[82]

VALVULAR DISEASES

Calcification of the mitral and/or aortic valve is four times more common in patients on dialysis than in matched controls and is associated with increased intima-media thickening and arterial calcification, perhaps suggesting a common pathogenic mechanism.[60] Consequences of valvular calcification include acceleration of aortic sclerosis—and possible progression to symptomatic aortic stenosis and consequent pressure overload of the left ventricle[83]—and incompetence of the mitral valve (exacerbated by left ventricular dilation), which, if present, further contributes to volume overload of the left ventricle.

DYSRHYTHMIA

Chronic kidney disease is associated with impaired intracardiac conduction, manifest as prolongation of the PQ and QRS intervals.[84] In critically ill patients[85] and infarct survivors[86] an eGFR less than 60 mL/min/1.73 m^2 has been consistently associated with two- to threefold increases in the risk of arrhythmias, such as atrial fibrillation, ventricular tachycardia, and ventricular fibrillation. The prevalence of arrhythmias is high among dialysis patients, with one study identifying nonsustained and sustained ventricular tachycardia in 25% and 57% of hemodialysis patients, respectively.[87] The contribution that these abnormalities make to the increased risk of sudden death among patients on dialysis and after transplantation is not known.

Most of the studies of electrophysiology in patients with CKD have been conducted in the hemodialysis population and therefore represent one extreme of the CKD phenotype. Hemodialysis is associated with particular risk factors that are known to trigger dysrhythmias, such as large changes in extravascular volume and rapid electrolyte shifts.[88] The excess of deaths around the first dialysis session of the week (i.e., after the typical longer interdialytic interval) also suggest these hemodialysis-specific factors are important[89] but perhaps not generalizable to other patients with CKD. However, other findings in this population do suggest that features associated with CKD per se, such as the presence of autonomic neuropathy (most marked among patients with diabetes) and reduced heart rate variability,[90,91] predispose to dysrhythmias.

Atrial fibrillation is also more common among patients with CKD.[92,93] The development of atrial fibrillation can reduce cardiac output in patients with left ventricular hypertrophy, who may rely on atrial contraction to fill the left ventricle. Furthermore, CKD is also a risk factor for thromboembolic complications of atrial fibrillation.[94] However, there is uncertainty about anticoagulation because the absolute excess of bleeding events is larger, so the risk-benefit balance is unproven (see discussion in Chapter 55). Furthermore, synthesis of matrix Gla protein (an important endogenous calcification inhibitor) is vitamin K–dependent; thus vitamin K antagonists may lead to excess vascular calcification.[93]

CLINICAL MANIFESTATIONS OF CARDIOVASCULAR DISEASE IN CHRONIC KIDNEY DISEASE

Whereas CAD explains over half of cardiovascular mortality in the general population,[95] studies have shown that CAD accounts for less than 20% of cardiovascular mortality in dialysis patients.[96–98] At some point therefore in the natural history of CKD, nonatherosclerotic cardiovascular disease evolves to become the dominant pathology.

As the GFR falls, there is an increasing burden of arterial stiffness and structural heart disease.[56,62,99] The clinical picture in patients with structural heart disease but normal kidney function includes heart failure, arrhythmias, and sudden cardiac death (SCD).[100] Similar syndromes are also observed in patients with advanced CKD. Data from the United States have suggested that the annual incidence of SCD in the CKD population is 2.8%, which is five times that in the general population.[101] In the dialysis population, SCD accounts for over half of all cardiovascular mortality.[101,102] Strategies that target nonatherosclerotic cardiac disease are required, and randomized trials of these strategies should be a high priority.

Although atherosclerotic disease may account for a smaller proportion of cardiovascular events among persons with advanced CKD, the absolute risk of atherosclerotic events is high: the risk of myocardial infarction among such patients is around 2% to 3% annually.[96,97] Strategies that target atherosclerotic disease may therefore be effective at reducing the burden of cardiovascular disease in those with advanced CKD.[101]

EPIDEMIOLOGY OF CARDIOVASCULAR DISEASE IN CARDIOVASCULAR DISEASE

Although the association between CKD and cardiovascular disease first became apparent from observations in young patients on dialysis, it is now recognized that the increased risk starts much earlier in the natural history of CKD.

ASSOCIATION BETWEEN KIDNEY FUNCTION AND CARDIOVASCULAR DISEASE

Numerous studies have shown that there is an inverse association between kidney function (at least as measured by the GFR or an estimate thereof) and cardiovascular risk.[8,104–110] Studies that have assessed the relationship between kidney function and cardiovascular risk generally fall into three categories—community-based prospective epidemiologic studies, observational data from randomized controlled trials, and analyses of health care management databases. Despite differences in the populations studied and adjustment for confounding variables, the results are surprisingly consistent. A metaanalysis of 19 creatinine-based studies (including over 160,000 events) has shown that for each 30% reduction in eGFR, the risk of major vascular events, which includes nonfatal and fatal events, increases by about 30%.[111]

The Chronic Kidney Disease Prognosis Consortium has combined the results of 21 studies in the general population.[9] After adjustment for age, gender, ethnicity, diabetes, blood pressure, total cholesterol, smoking, and history of cardiovascular disease, a lower eGFR was associated with an increased risk of death from any cardiovascular cause as compared with the reference group (eGFR, 90–104 mL/min/1.73 m^2; Fig. 54.4).

However, very few of the available studies have sought to assess the nature of any associations between reduced eGFR and particular types of cardiac events, such as coronary artery disease, heart failure, and cardiac arrhythmia. A community-based prospective study from Iceland has demonstrated an inverse relationship between eGFR and coronary artery disease specifically, but few participants had CKD and, if present, it was mild (mean eGFR in the CKD group, 58.7 mL/min/1.73 m^2).[110] Such information will be essential if we are to develop a better understanding of the reasons why reduced kidney function is associated with an excess risk of the aggregate of deaths from any cardiovascular cause.

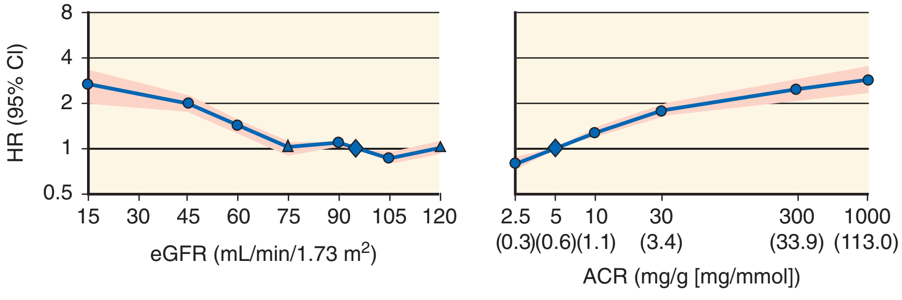

Fig. 54.4 Association between the estimated glomerular filtration rate (eGFR; *left panel*) and albuminuria *(right panel)* and cardiovascular mortality. This meta analysis included data from 1,234,182 participants in 21 cohorts from the general population. The albumin-to-creatinine ratio (ACR) was available from 14 studies of 105,872 participants.[9]

ASSOCIATION BETWEEN ALBUMINURIA AND CARDIOVASCULAR DISEASE

Data from the Chronic Kidney Disease Prognosis Consortium have indicated that among persons without known kidney disease, a higher level of albuminuria is associated with an increased risk of cardiovascular disease, and there is no apparent threshold below which lower albuminuria is not associated with lower risk (see Fig. 54.4).[9] In another meta-analysis of 26 cohorts with over 7000 coronary events, there was a continuous association between the degree of albuminuria and risk of coronary artery disease.[112] Compared with "normal" levels of albuminuria (i.e., <2.5 mg/mmol in men or <3.5 mg/mmol in women), microalbuminuria (i.e., <30 mg/mmol) was associated with a 50% increase in risk of coronary artery disease (HR, 1.47; 95.% CI, 1.30–1.66), and macroalbuminuria (≥30 mg/mmol) was independently associated with a doubling of risk (HR, 2.17; 95% CI, 1.87–2.52). The association between albuminuria and cardiovascular risk appears to be independent of the GFR.[9,10,113]

Because leakage of albumin into the extravascular space may occur as a result of endothelial dysfunction, it has been suggested that some albuminuria may represent a renal manifestation of progressive, diffuse arterial disease.[114,115] If true, this could imply that some or all of the observed association between albuminuria and cardiovascular disease is attributable to residual "confounding by disease," in which unmeasured vascular disease (manifest as endothelial dysfunction) is both a cause of albuminuria and of cardiovascular events. Because available treatments that reduce albumin leakage also exert other salutary effects on cardiovascular risk, it is unlikely that the multiple benefits of these therapies could be disentangled. It therefore remains unclear whether reducing albuminuria per se will reduce the risk of vascular disease. Indeed, trials of more -intensive versus less intensive albuminuria reduction strategies (e.g., adding angiotensin-converting enzyme or direct renin inhibitors to angiotensin receptor blockers) have not shown additional cardiovascular protection, although their statistical power to do so was limited.[116,117]

KIDNEY DISEASE AS A CAUSE OF CARDIOVASCULAR DISEASE

As described in earlier chapters, the kidneys are responsible for a wide variety of physiologic processes, including clearance of metabolic waste products, salt and water balance, blood pressure regulation, and hormone production. Even mild kidney disease has direct effects on the regulation of these processes and could therefore potentially lead to disturbed function and metabolism in other systems, including the cardiovascular system. It is of considerable interest therefore to consider to what extent the excess risk of cardiovascular disease in CKD could be explained by such disturbances.

In this section and later (Indirect Risk Factors: Causes of Both Kidney and Cardiovascular Disease) we suggest a framework for considering the evolution of cardiovascular risk in CKD, as illustrated in Fig. 54.5. We consider two main types of risk factors:

1. Direct risk factors (e.g., hypertension) that arise as a direct consequence of kidney damage and are associated with one or more types of cardiovascular disease (Table 54.1)
2. Indirect risk factors (e.g., diabetes mellitus, obesity) that cause both kidney disease and one or more types of cardiovascular disease (see later)

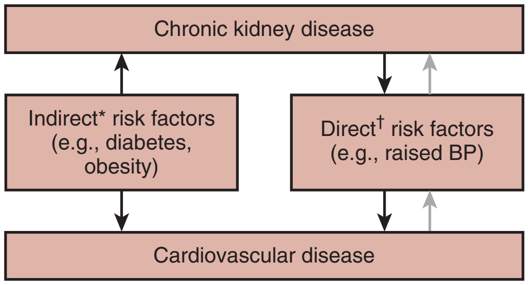

Fig. 54.5 The relationship between chronic kidney disease (CKD) and cardiovascular disease is mediated by direct and indirect risk factors. *Indirect risk factors are those that cause both renal disease and one or more types of cardiovascular disease; †Direct risk factors are those that arise as a direct consequence of kidney damage and are associated with one or more types of cardiovascular disease.

In general, observational studies of particular risk factors among persons with kidney disease may not provide quantitatively reliable information about any associations that might exist because of the difficulty of adjusting for confounding by disease, also known as "reverse causality."[15] For example, reduced left ventricular function in patients on dialysis may lead to lower blood pressure, yielding an apparent association between lower blood pressure and cardiovascular mortality, which is not one of cause and effect.[15,118] Paradoxically therefore it may be more appropriate to rely chiefly on epidemiologic studies performed in nonrenal populations when assessing the potential relevance of particular risk factors to the cause of the various types of cardiovascular disease observed in CKD.

BLOOD PRESSURE

The kidney is centrally involved in blood pressure regulation. Even relatively mild kidney damage can raise blood pressure, mediated by salt and water retention (and hence intravascular volume expansion), sympathetic overactivity, activation of the renin-angiotensin system, and accumulation of endogenous vasopressors.[119] In turn, hypertension can damage the kidneys further, leading to a vicious cycle of rising blood pressure (with arterial hyalinosis and vascular stiffening) and declining GFR. Evidence from kidney donors, who are selected for being healthy and in particular having preserved kidney function, have suggested that a 10-mL/min/1.73 m^2 reduction in GFR leads directly (i.e., causally) to a 5-mm Hg increase in systolic blood pressure. However, this may be an underestimate because elevations in blood pressure are likely to be treated in this population.[113]

Hypertension is strongly associated with several different types of cardiovascular disease in the general population. Prospective epidemiologic studies have indicated that there is a log-linear association between higher blood pressure and an increased risk of coronary artery disease, ischemic and hemorrhagic stroke, and congestive heart failure. The Prospective Studies Collaboration, in a meta analysis of 61 prospective studies (which included a total of 1,000,000 adults without prior cardiovascular disease and 56,000 cardiovascular

Table 54.1 Direct Risk Factors Linking Chronic Kidney Disease and Cardiovascular Disease[a]

Risk Factor	Estimated Change (for each 10-mL/min/1.73 m² lower eGFR[b])	Approximate Stage of Chronic Kidney Disease at Which Abnormality Manifests	Approximate Relative Risk[c]	Causal?	Comments
Coronary Artery Disease					
Systolic blood pressure	↑5 mm Hg	1–2	1.2	Yes	Reliable data from large-scale observational[120] and RCTs[121] support a causal relationship.
HDL cholesterol	↓0.2–0.4 mmol/L	3	1.2	Possible	Large-scale observational studies support an inverse association with coronary disease.[128] Genetic studies and randomized trials are inconclusive.
Lipoprotein(a)	↑0.2–0.4 µmol/L	3	<1.5	Possible	Genetic studies support a causal association with coronary disease and aortic valve disease.[138,203] Most of excess risk is in the top fifth of lipoprotein(a) distribution—that is, individuals with smaller apolipoprotein(a) particles; requires confirmation in trials.
Triglycerides	↑0.1 mmol/L	3	1.1	Possible	Classical observational studies are unclear, but genetic data support a causal association; requires confirmation in trials.
Phosphate	↑0.3 mmol/L	3b–4	1.4	Possible	Observational studies in general population[170] suggest possible causal link; requires confirmation in trials.
PTH	↑3 pmol/L	3	1.3	Possible	Observational studies in general population suggest possible causal link.[190,204]
Anemia	↓0.2–0.5 g/dL	3b–4	—	No	RCTs in CKD population do not show that correcting anemia improves cardiac outcomes.[205]
Homocysteine	↑0.5 µmol/L	3	<1.5	No	RCTs do not demonstrate reduced CHD risk with reduction in homocysteine.[161]
Uric acid	10-15 µmol/L	2–3	1.1	No	Mendelian randomization studies suggest observed associations are probably confounded.[202]
Congestive Heart Failure					
Systolic blood pressure	↑5 mm Hg	1	1.2	Yes	Reliable data from large-scale observational[120] and RCTs[121] support causal relationship.
Phosphate	↑0.3 mmol/L	3b–4	1.4	Possible	Observational studies in general population[170] suggest possible causal link; requires confirmation in trials.
PTH	↑3 pmol/L	3	1.3	Possible	Observational studies in general population (and secondary outcomes of trials) suggest possible causal link.[190,204,206]
Anemia	↓0.2–0.5 g/dL	3b–4	—	No	Although observational data support link between anemia and structural heart disease,[147] randomized trials do not show that correcting anemia improves cardiac outcomes.[205]
FGF23	↑10 RU/mL (varies across range of GFR)	1–2	<1.5	Possible	Supportive experimental and observational data in CKD and general population.[180]
Stroke					
Systolic blood pressure	↑5 mm Hg	1–2	1.3	Yes	Reliable data from large-scale observational[120] and RCTs[121] support causal relationship.
HDL cholesterol	↓0.2–0.4 mmol/L	3	1.1	Possible	Weak inverse association in observational studies.[128]
Phosphate	↑0.3 mmol/L	3b–4	1.4	Possible	Observational studies in general population[170] suggest possible causal link.

Table 54.1 Direct Risk Factors Linking Chronic Kidney Disease and Cardiovascular Disease[a] (Cont'd)

Risk Factor	Estimated Change (for each 10-mL/min/1.73 m^2 lower eGFR[b])	Approximate Stage of Chronic Kidney Disease at Which Abnormality Manifests	Approximate Relative Risk[c]	Causal?	Comments
Anemia	↓0.2–0.5 g/dL	3b–4	—	No	No supportive observational data in general population.[207] RCTs in CKD population suggest that correcting anemia with ESA may cause stroke.[208]
Homocysteine	↑0.5 μmol/L	3	<1.5	No	RCTs do not demonstrate reduced stroke risk with reduction in homocysteine.[161]

[a]The table shows typical differences in risk factors for coronary artery disease, stroke, and congestive heart failure between patients with mild renal impairment and middle-aged healthy general population.
[b]Associated with change in risk factor induced by a 10-mL/min reduction in the GFR.
[c]Associated with change in risk factor induced by 10 mL/min reduction in GFR.
CHD, Coronary heart disease; *eGFR*, estimated glomerular filtration rate; *ESA*, erythropoiesis-stimulating agent; *PTH*, parathyroid hormone; *RCT*, randomized controlled trial.

deaths), has shown that a prolonged 20-mm Hg increment in usual systolic blood pressure is associated with a more than twofold higher risk of stroke-related death and a twofold higher risk of death due to coronary artery disease and of death from heart failure.[120]

That these associations are causal has been established reliably by randomized trials showing that lowering systolic blood pressure reduces the risk of stroke, coronary artery disease, and of other cardiovascular diseases, with every 5-mm Hg reduction associated with a reduction of about 29% in risk.[121] The results are broadly similar for different classes of antihypertensive treatment, although there is some evidence to suggest that inhibitors of the renin-angiotensin system may have particularly beneficial effects on the risk of heart failure.[122]

Because hypertension appears very early in the natural history of CKD, a patient who has progressed to stage 3 CKD will generally have been exposed to a prolonged increase in blood pressure, which, depending on the degree of treatment received, would be expected to contribute substantially to the subsequent risk of nonatherosclerotic and atherosclerotic cardiovascular disease. For example, a 5-mm Hg difference resulting from a mild reduction in GFR (e.g., from 90 to 80 mL/min/1.73 m^2) could translate into an increase of about 25% for the risk of stroke death and 20% for the risks of coronary artery disease–associated death and other cardiovascular death. These relative risks may be substantially larger in younger individuals, in whom the association between blood pressure and cardiovascular death is stronger than in older individuals.[120]

DYSLIPIDEMIA

The characteristic lipid profile resulting from impaired kidney function beyond about stage 3 CKD is an accumulation of partially catabolized, triglyceride-rich, very-low-density lipoprotein (VLDL) and intermediate-density lipoprotein (IDL) particles, leading to an elevated serum triglyceride concentration, with lower high-density-lipoprotein (HDL) cholesterol concentrations. LDL cholesterol concentration in CKD is similar or lower than the population average, except that nephrotic-range proteinuria leads to an increase in LDL cholesterol.[123] Cholesterol continues to be progressively removed from these particles, and triacylglycerol is added, leading to an excess of small, dense, LDL particles, which may be more atherogenic.[124] HDL function may also be impaired in CKD.[125] In addition, lipoprotein(a)—Lp(a)—concentrations are increased in association with CKD; again, this abnormality is a direct result of impaired kidney function because it can be normalized by kidney transplantation.[126,127]

Based on what is known from epidemiologic studies in persons without CKD, it remains unclear whether the typical dyslipidemia seen in CKD would be expected to result in a large increase in the risk of atherosclerotic events in patients with (nonnephrotic) CKD. Although it is clear from epidemiologic studies[95,128] and large-scale randomized trials[129] that elevated LDL cholesterol is a contributing cause of atherosclerotic events, the observed incidence of atherosclerotic events in CKD cannot be caused chiefly by increased LDL cholesterol concentrations because such an abnormality occurs in a minority of patients, usually in association with severe proteinuria.

The relevance of the increased triglyceride concentrations is unclear. Traditional observational epidemiologic studies have suggested that the positive association between triglyceride concentration and cardiovascular risk is confounded and does not persist once adjustment is made for HDL cholesterol and other confounders.[128] However, more recent genetic epidemiologic studies have shown that single-nucleotide polymorphisms (SNPs) associated with higher triglyceride concentrations are also associated with higher cardiovascular risk, and that this association is attenuated by adjustment for triglyceride concentration, suggesting that triglycerides are causally associated with cardiovascular risk.[130] In cross-sectional studies, triglyceride concentrations increase as the GFR falls, so that a 10-mL/min/1.73 m^2 reduction in the eGFR is associated with about a 0.1-mmol/L increase in triglyceride concentration.[131]

HDL cholesterol is generally reduced in CKD. In one study, for example, patients on hemodialysis had a mean HDL cholesterol of 0.89 mmol/L compared with 1.4 mmol/L in healthy controls.[132] Prospective studies have shown that a 0.4-mmol/L (15-mg/dL) higher HDL cholesterol is associated with a 20% reduction in coronary events (HR, 0.78; 95% CI, 0.74–0.82),[128] whereas the association with ischemic stroke is less clear.[128,133] However, trials of drugs that can raise HDL cholesterol have so far not shown clear benefit, and genetic studies have also been inconclusive.[134–137] Thus it remains unclear whether reduced HDL cholesterol is a cause of coronary heart disease (CHD) in the general or CKD population.

It is also possible that an elevated Lp(a) might contribute to an increase in risk of CHD. Modest reductions in the GFR are associated with an increase of 0.2 to 0.4 μmol/L in Lp(a) concentrations,[126] and genetic data have strongly suggested that there is a causal association between Lp(a) concentration and coronary artery disease in the general population.[138] However, most of the excess risk attributable to Lp(a) occurs among individuals with smaller apolipoprotein(a)—apo(a)—moieties, which is largely genetically determined by the number of kringle(IV) repeats,[139] whereas most of the increase in Lp(a) among patients with CKD occurs in patients with larger apo(a) moieties.[126] It is therefore uncertain how much of the excess risk of CHD observed in CKD can be attributed to altered Lp(a) metabolism.[140]

In summary, it is unclear to what extent the dyslipidemia typically found in association with CKD contributes materially to the excess risk of atherosclerotic cardiovascular disease observed in CKD. However, in the general population, reducing LDL cholesterol reduces the risk of atherosclerotic events, even among people with average or low LDL cholesterol values,[129] and this strategy has also proven effective in reducing atherosclerotic risk in CKD (see "Cardiovascular risk prevention: Dislipidemia").

OTHER DIRECT RISK FACTORS

In addition to increased blood pressure, which has an established association with cardiovascular disease in the general population, several other metabolic disturbances caused by kidney disease may contribute to some of the increased risk of cardiovascular disease.

COAGULATION DEFECTS

Chronic kidney disease is associated with raised fibrinogen concentrations, which in turn would increase plasma viscosity and modulate coagulation in a procoagulant direction.[141,142] Chronic kidney disease also increases factor VIII and von Willebrand factor concentrations.[143] Although increases in fibrinogen are not well characterized in CKD, a 1-g/L increase in fibrinogen concentration has been reported in one study.[143] Epidemiologic studies in the general population have suggested that after adjustment for potential confounders, such an increase is associated with a 1.8-fold excess risk of coronary disease, stroke, and other cardiovascular events.[144] However, the lack of an association between genetic variants that determine fibrinogen concentrations and cardiovascular risk make it less plausible that raised fibrinogen itself is a cause of atherosclerotic disease, so the relevance of any such increase in CKD is uncertain.[145]

ANEMIA

Anemia in CKD is caused by a combination of factors, which include erythropoietin deficiency, functional iron deficiency, and chronic inflammation (see Chapter 55). The effects of chronic anemia on the cardiovascular system are not well studied in the general population because anemia is generally associated with disease. In CKD, however, anemia is associated with left ventricular hypertrophy, in one study, a 0.5-g/dL lower hemoglobin concentration was associated with a 30% higher frequency of increased left ventricular mass (defined as 20% greater than baseline).[146] In another study of dialysis patients, each 1-g/dL decrease in hemoglobin was associated with a 50% increased risk of left ventricular dilation and a 25% increased risk of cardiac failure.[147]

Although some small nonrandomized studies have suggested that partial correction of anemia reduces left ventricular mass index in patients with CKD,[148,149] randomized trials have not shown that complete correction of anemias, compared with partial correction, leads to a reduction in left ventricular mass.[150–152] It is therefore unclear whether anemia is a cause of structural heart disease in CKD but, because it does not appear until a significant amount of kidney function is lost (e.g., eGFR < ≈40 mL/min/1.73 m^2),[153] it seems unlikely that it could contribute substantially to the excess risk of cardiovascular disease observed among those with a higher eGFR. The results of randomized trials of treatments to increase hemoglobin that involved assessment of cardiovascular endpoints are discussed later ("Cardiovascular risk prevention: Anemia") and reviewed in the context of treatments of anemia in Chapter 55.

HOMOCYSTEINE

The observation that homocystinuria, which results in very large increase in plasma homocysteine concentrations is associated with premature cardiovascular disease led to interest in whether more moderate increases in homocysteine might be associated with an increased risk of coronary artery disease.[154] Homocysteine concentrations have a strong inverse association with the GFR and, because they are reduced by kidney transplantation, the rise in the homocysteine level is a direct result of impaired kidney function.[155] However, the mechanism(s) underlying hyperhomocysteinemia in CKD are complex and not fully understood. They are not attributable to reduced clearance, because renal homocysteine excretion accounts for less than 1% of its elimination, so it must instead involve metabolic disturbances in remethylation and transsulfuration pathways.[156] Moderate reductions in the GFR are associated with a 5-μmol/L increase in total plasma homocysteine.[157]

A meta analysis of prospective observational studies has shown that a 25% lower usual homocysteine concentration (≈3 μmol/L) is associated with an 11% lower risk of coronary disease and a 19% lower risk of stroke.[158] The earliest genetic studies of *MTHFR* mutations appeared consistent with this association being causal,[159] but subsequent metaanalysis of all available genetic data have indicated that such an association is unlikely to reflect causality.[160] Large-scale randomized trials of folate and B vitamins, which reduce plasma homocysteine concentration, have also failed to demonstrate a benefit of lowering homocysteine (see later, "Cardiovascular risk prevention: Homocysteine").[161–163]

CHRONIC KIDNEY DISEASE AND MINERAL BONE DISORDERS

In regard to chronic kidney disease and mineral bone disorders (CKD-MBD), impaired kidney function causes a complex disorder of calcium and phosphate metabolism, including effects on vitamin D and parathyroid hormone (see Chapter 53 for a detailed description), which become especially marked after stage 3b CKD. Because such abnormalities are uncommon in persons without CKD, the information we have available is therefore derived mostly from observational studies conducted in patients with CKD, which, because of the likelihood of residual confounding, severely limits their interpretation.

Phosphate

Serum phosphate concentrations rise once the kidney's ability to excrete the dietary phosphate load is exceeded.[153] Multiple homeostatic processes are in place to maintain serum phosphorus concentrations in the normal range, despite diminished excretion with impaired kidney function, and hence a serum phosphate concentration within the reference range does not necessarily imply normal phosphate metabolism.[164] Higher serum phosphate concentrations are associated with increased arterial stiffness[165,166] and calcification,[167,168] perhaps because hyperphosphatemia can induce vascular smooth muscle cells to develop an osteoblastic phenotype.[169] Such arterial stiffness could plausibly increase the risk of structural heart disease and thereby of heart failure and cardiac arrhythmias[76]

At least within the range of concentrations observed in the general population, there is an association between higher serum phosphate levels and increased risk of cardiovascular events and mortality.[170,171] In the Framingham Offspring Study, for example, a 1-mg/dL (0.32-mmol/L) increment in the serum phosphate level was associated with an increase in the risk of cardiovascular disease—namely, coronary artery disease, stroke, peripheral arterial disease, or heart failure—of about one-third (HR, 1.30; 95% CI, 1.05–1.63) after adjustment for conventional cardiovascular risk factors and the eGFR.[170] However, there were too few events to allow the association with heart failure to be assessed separately from typical atherosclerosis-related events.

Randomized trials of treatments that lower serum phosphate levels (e.g., phosphate binders) are required to determine whether hyperphosphatemia, or the counterregulatory processes that are activated in CKD, are causally related to cardiovascular disease. The issue is complicated by the possibility of adverse effects of calcium-containing phosphate binders; these may contribute to vascular calcification by inducing a positive calcium balance, which results in the deposition of calcium in the vasculature rather than bone.[172]

Fibroblast Growth Factor 23

FGF23 concentrations rise earlier in CKD than the serum concentrations of phosphate or even parathyroid hormone[173] and help maintain a normal serum phosphate concentration via a phosphaturic effect. The association of FGF23 with mortality was first demonstrated in patients on maintenance hemodialysis[174] and, since then, other studies have found associations between FGF23 concentrations and all-cause and cardiovascular mortality in patients with CKD.[175–177] More recently, FGF23 was studied in a population without prior cardiovascular disease and over 10 years of follow-up. In this study, it was again independently associated with all-cause mortality (HR, per doubling 1.25; 95% CI, 1.14–1.36) and incident heart failure events (697 events; HR, 1.41; 95% CI, 1.23–1.61) but not incident atherosclerotic events (797 events; HR, 1.12; 95% CI, 0.98–1.29).[178]

Unlike phosphate, serum FGF23 concentrations do not correlate with arterial calcification.[179] Experimentally, FGF23 has been shown to induce left ventricular hypertrophy in the absence of its coreceptor klotho; furthermore, FGF23 and left ventricular mass are positively associated among patients with CKD and those with normal renal function.[180,181] Assuming that these associations are causal, the biologic mechanisms linking FGF23 with cardiovascular diseases may be different from those for serum phosphate.

Vitamin D

As discussed in more detail in Chapter 53, vitamin D is activated (via 1α-hydroxylation) by the kidney, and therefore fully active 1,25-dihydroxy vitamin D concentrations decline early in the progression of CKD.[153,182] In addition, rising FGF23 concentrations also inhibit the activation of vitamin D. There is also an association between "native" vitamin D (usually measured as 25-hydroxy vitamin D) concentration and the GFR.[182] However, this may reflect poorer nutrition in patients with chronic disease. Vitamin D deficiency (and subsequent hypocalcaemia) removes negative feedback from the parathyroid glands, and parathyroid hormone (PTH) concentrations therefore rise as GFR falls.[153]

Some observational studies in the general population have suggested that there is an inverse association between 25-hydroxy vitamin D concentrations and subsequent cardiovascular events, whereas others have not.[183] A metaanalysis has demonstrated inverse associations between 25-hydroxy vitamin D and all-cause mortality (and its components including vascular, cancer and other non-vascular causes).[184] Randomized trials of vitamin D supplementation in the general population have not demonstrated a reduction in cardiovascular events.[183] However, it is possible that these trials used doses of vitamin D that were too low to induce a potent enough effect. Some studies have suggested that vitamin D deficiency predisposes patients to diabetes mellitus and raises blood pressure, but again these data are not conclusive.[183] It is unclear whether disturbed vitamin D metabolism could plausibly contribute to the observed excess risk of cardiovascular disease in CKD.

Parathyroid Hormone

PTH concentrations rise as a direct result of declining GFR early in the progression of CKD; secondary hyperparathyroidism (sHPT) can be found in 20% of patients with CKD stages 1 and 2.[153] PTH rises because of a lack of negative feedback from declining 1,25-hydroxy vitamin D and calcium concentrations and rising concentrations of serum phosphate, even within the population reference range. Parathyroid hormone has been implicated in atherogenesis (and in calcification of atherosclerotic lesions) and also in modifying cardiac fibrosis in animal models.[185,186] There have been case reports of improved myocardial metabolism and structure after parathyroidectomy, suggesting that high concentrations of PTH may damage the heart.[187] There does not appear to be a clear association between PTH and all-cause and cardiovascular mortality in patients with CKD,[188] although one study has reported an association between elevated PTH and all-cause mortality in a cohort of persons with CKD stage 3.[189] Studies in the general population, which may be less prone to confounding, have suggested that there is an independent association between elevated PTH concentrations and fatal and nonfatal cardiovascular disease.[190] A metaanalysis of 12 such studies has shown that the risk of cardiovascular disease is 50% (HR, 1.50; 95% CI, 1.18–1.92) higher among patients in the groups with highest PTH concentration compared with those in the lowest.[190]

OXIDATIVE STRESS AND INFLAMMATION

Oxidative stress is defined as tissue injury resulting from an excess of oxidant compounds; it is important in protecting against infection and tissue repair. Oxidative stress leads to a reduction in nitric oxide bioavailability and thus to endothelial dysfunction[191] and may also contribute to left ventricular remodeling and fibrosis and oxidation of lipoproteins.[192] Chronic kidney disease is associated with oxidative stress,[193] although precise quantification is difficult. In the general population, dietary intake or plasma concentrations of antioxidant vitamins (vitamin A [beta-carotene], C, and E) are inversely related to cardiovascular disease incidence and mortality,[194] but randomized trials of antioxidant treatments have indicated that at least in the doses studied, use of these drugs does not reduce the risk of cardiovascular disease.[195]

Patients with CKD have increased measures of inflammation compared with age- and gender-matched controls.[141] Genetic data from the general population have suggested that inflammation is causally associated with coronary disease, although the mechanism remains unclear.[196,197] A recently completed trial of canakinumab (an interleukin-1 β [IL-1β] antagonist) in persons with previous myocardial infarction and high-sensitivity C-reactive protein (hs-CRP) greater than 2 mg/L has reported a 15% relative risk reduction (HR, 0.85; 95% CI, 0.74–0.98; $P = .021$)) in nonfatal myocardial infarction or stroke or cardiovascular death observed in the group receiving a dose of 150 mg every 3 months, supporting the hypothesis that inflammation mediated via IL-1β does contribute to atherosclerotic cardiovascular events.[198,199]

URIC ACID

Uric acid is cleared by the kidney, and its concentration rises as GFR falls (so that each 10- mL/min/1.73 m^2 lower eGFR is associated with a 10–15 μmol/L increase in uric acid concentration). Positive associations between uric acid concentration and the risk of CHD have been reported, which would suggest that a 10- to 15-μmol/L increase in uric acid would increase the risk of CHD by about 10%.[200,201] However, mendelian randomization experiments that took account of the pleiotropic effects of genetic variants have cast doubt on whether such associations are causal.[202]

INDIRECT RISK FACTORS: CAUSES OF KIDNEY AND CARDIOVASCULAR DISEASE

DIABETES MELLITUS

Diabetes mellitus is a common disease that affects about 150 million people worldwide. The overall prevalence is increasing—driven in part by a rise in the prevalence of obesity and an aging population—and it has been estimated that there will be over 300 million persons globally with type 2 diabetes by 2025.[209] A metaanalysis of 23 studies, including 21, 237 cardiovascular deaths, has found that the risk of cardiovascular death at a given eGFR or urine albumin-to-creatinine ration (UACR) was 1.2 to 1.9 times higher among individuals with diabetes than in those without, with no evidence of an interaction.[210]

Diabetic nephropathy is caused by the hemodynamic effects of chronic hyperglycemia and glycation of structural proteins within the kidney, beginning with the glomerular basement membrane but later causing mesangial and interstitial matrix expansion (see Chapter 39). Approximately 40% of patients new to dialysis in the United States have diabetic nephropathy, and diabetes mellitus is present as a comorbid illness in an additional 10% of patients; in other words, diabetes mellitus is present but is thought not to be the primary cause of ESKD.[101] Strict glycemic control and blood pressure control (with renin-angiotensin system inhibitors) both appear to retard the progression (and perhaps the development) of diabetic nephropathy.[211–215] Patients with diabetes (type 1 or 2) are at increased risk of mortality, and about two-thirds of deaths can be attributed to cardiovascular causes.[216]

Prospective data from a meta analysis of 102 studies with information on fatal and nonfatal outcomes have demonstrated that the presence of diabetes mellitus doubles the risk for coronary heart disease (HR, 2.00; 95% CI, 1.83–2.19).[217] The risk of ischemic stroke was increased by about half (HR, 1.56; 95% CI, 1.56–2.09), hemorrhagic stroke by around four-fifths (HR, 1.84; 95% CI, 1.54–2.13), and other vascular deaths (including heart failure) by about three-quarters (HR, 1.73; 95% CI, 1.51–1.98). These estimates were not significantly affected by adjustment for traditional cardiovascular risk factors (e.g., lipid levels and blood pressure), but were also unaffected by adjustment for kidney function.[217] Among patients with CKD therefore the presence of diabetes mellitus seems likely to contribute additional risk over and above any risk arising from direct risk factors associated with impaired kidney function (see Fig. 54.5).

OBESITY

In the general population, observational studies have shown that obesity is associated with an increased risk of cardiovascular disease above a body mass index (BMI) of 25 kg/m^2, with each 5 kg/m^2 associated with an increase in risk of cardiovascular mortality of around 40% (HR, 1.41; 95% CI, 1.37–1.45).[218] This association is probably causal and mediated by the known adverse effects of adiposity on blood pressure, lipoprotein levels, and glucose intolerance.

Obesity is also known to increase the risk of death from nonneoplastic kidney disease by about 60% (HR, 1.59; 95% CI, 1.27–1.99) per 5 kg/m^2 above a BMI of 25 kg/m^2.[218] A systematic review has indicated that as compared with a BMI of 18 to 25 kg/m^2, being overweight (BMI ≥ 25 but < 30 kg/m^2) is associated with a 40% (HR, 1.40; 95% CI, 1.30–1.50) increase in the risk of kidney disease, whereas being obese (BMI ≥ 30 kg/m^2) is associated with an 80% increased risk (HR, 1.83; 95% CI, 1.57–2.13).[219] This association may be partly mediated by effects on blood pressure and diabetes, but these do not explain all the excess risk observed.[220] Kidney biopsies of overweight individuals (without diabetes or hypertension) may show typical features of glomerular hyperfiltration, a known mechanism of premature decline in kidney function.[221] Furthermore, weight loss can reduce blood pressure and proteinuria.[222]

CARDIOVASCULAR RISK PREDICTION

Risk prediction models in the CKD population are desirable because they may help target therapies for those most at risk. Most people with stage 4 or 5 CKD are at high risk of atherosclerotic disease and so would be suitable candidates for effective treatments, but at earlier stages of CKD, the risk of cardiovascular events may be lower, so models are needed to assess who should be treated. Models that involve demographic variables (e.g., age, gender), basic clinical measurements (e.g., blood pressure, anthropometric measurements), and simple blood and urine biomarkers (e.g., lipid fractions, albuminuria) have now been developed, and there are also versions that incorporate imaging or other measurements (e.g., pulse wave velocity).

BASIC RISK PREDICTION SCORES

Equations have been developed in the general population that can be used to predict cardiovascular events based on knowledge of a few basic parameters. For example, the Framingham risk score requires knowledge of the patient's age, gender, smoking status, total and HDL cholesterol, and systolic blood pressure.[223,224] However, such equations are inaccurate and generally underestimate the incidence of cardiovascular events in patients with CKD.[225]

ROLE OF ESTIMATED GFR IN RISK PREDICTION

Estimated GFR equations were developed to assess kidney function without the need for invasive direct measurements. Since their introduction, the association between eGFR and cardiovascular risk has been the subject of much research. The most widely used filtration marker is creatinine, but its inadequacies in terms of measuring kidney function are well-recognized (including dependence on muscle mass and significant tubular secretion at low GFR). However, nonrenal determinants of creatinine and other filtration markers such as cystatin C may improve cardiovascular risk prediction[226] because they incorporate information on other cardiovascular risk factors such as diabetes mellitus.[227] Indeed, combining filtration markers may lead to better risk prediction.[228]

USE OF IMAGING AND OTHER ASSESSMENTS OF CARDIOVASCULAR FUNCTION IN RISK PREDICTION

Blood markers of myocardial damage and dysfunction, such as cardiac troponin and brain natriuretic peptide (B type), strongly predict cardiovascular risk among patients with CKD,[229] including those on dialysis.[230,231] Furthermore, numerous studies have demonstrated that echocardiographic measures can provide useful prognostic information. This includes both static measures (e.g., left ventricular mass) and dynamic measures (e.g., dobutamine stress echocardiography). For example, left ventricular mass predicts mortality in patients on dialysis,[72,232–234] and the presence of systolic dysfunction can provide further information.[235] Echocardiography is therefore recommended for routine use in patients on dialysis,[236] but its additional value over more simple risk stratification tools has not been studied in detail.

CIMT has been used in the general population to predict the risk of cardiovascular events, but variation in the technique used makes comparison of studies difficult. Although CIMT appears to predict the risk of coronary and cerebrovascular events in the general population, it does not appear to provide additional prognostic information to a standard score, such as that derived from the Framingham equation.[237] In the CKD population, it is unclear whether CIMT provides independent prognostic information; one study of 203 Chinese patients with stage 3 or 4 CKD has found that CIMT is independently associated with outcome,[238] whereas another study of 315 stages 4 or 5 CKD patients did not find this relation.[34]

Vascular calcification can also be assessed using x-ray–based methods, and its presence is independently associated with subsequent cardiovascular events.[239,240] However, in the general population, the value of coronary calcification scores is uncertain because it may not add useful information to traditional scores.[241] The incremental predictive value of arterial calcium scoring has not been studied in detail in CKD populations. Similarly, the functional equivalent of calcification (i.e., arterial stiffness) independently predicts the risk of subsequent cardiovascular events in some studies,[34] but the method used may be important (see earlier, "Arterial Stiffening").

The purpose of risk prediction is to identify a high-risk group of patients who will benefit most from interventions. It is important to note that treatment decisions should not be based solely on the risk factor that will be modified by the treatment in question. For example, decisions on whether to lower LDL cholesterol should not be based on measured LDL cholesterol, but rather should be a holistic risk assessment that considers all available data. Treatment decisions based on single risk factors would lead to the undertreatment of the highest-risk patients. The next section will discuss the available evidence from randomized trials of treatments for cardiovascular disease in the CKD population.

CARDIOVASCULAR RISK PREVENTION

In general, individual treatments for the prevention of cardiovascular events are at best only moderately effective, yielding relative risk reductions of at most 25%, so the detection of such effects can only be reliably achieved through large-scale randomized trials.[242] Observational studies are prone to at least moderate biases that may obscure or mimic treatment effects.[243] Because the pathophysiology of cardiovascular disease may be qualitatively different once patients reach stages 3 to 5 CKD, with a higher proportion of events attributable to structural heart disease, extrapolation of the results of trials conducted in the general population may not be appropriate. It is necessary therefore to consider what information is available from subgroup analyses of large trials, as well as from trials conducted exclusively among patients with CKD, before determining whether to extrapolate results obtained in studies in the general population to patients in varying stages of CKD.

SMOKING CESSATION

In the general population, cigarette smoking is associated with an increased risk of cardiovascular disease,[244] and the beneficial effects of smoking cessation provide strong evidence that the association is one of cause and effect.[224,245] Substantial numbers of patients with CKD smoke tobacco,[246,247] and recent data have demonstrated that the hazards are similar in patients with CKD to those in the general population.[248] As a consequence, the absolute excess cardiovascular (and noncardiovascular) risks attributable to smoking are likely to be greater in smokers with CKD than in similarly aged smokers without CKD, and therefore the potential benefits of cessation are substantial.

BLOOD PRESSURE

Among persons without CKD, randomized trials have shown clearly that lowering blood pressure reduces the risk of subsequent cardiovascular events.[121] Although there may be subtle differences in efficacy among different antihypertensive regimens, the major determinant of benefit is the absolute magnitude of any blood pressure reduction that is achieved.[122] After standardizing for the amount of blood pressure reduction, the relative reduction in risk appears to be approximately independent of the initial blood pressure level. This suggests that antihypertensive treatments will reduce the risk of cardiovascular events, even in those without obviously elevated blood pressure, but in those who are at increased risk for other reasons.[249]

Information about the effects of lowering blood pressure among persons with CKD stages 1 to 3 is available from trials conducted largely in people without known kidney disease. For example, the Perindopril Protection Against Recurrent Stroke (PROGRESS) study included 1757 participants with stage 3 or higher CKD among the 6105 participants with prior cerebrovascular disease who had been enrolled.[250] Assignment to active therapy was associated with a 35% reduction in the risk of stroke in subjects with CKD (HR, 0.65; 95% CI, 0.50–0.83), which was similar to the risk reduction observed in the whole study population. Because subjects with CKD had a higher background risk of cardiovascular events, the absolute treatment effect was 1.7-fold greater than for patients without CKD.[251]

Similarly, in a posthoc analysis of the Heart Outcomes and Prevention Evaluation (HOPE) study, 980 patients with impaired kidney function (serum creatinine level >124 μmol/L [1.4 mg/dL]) were studied,[252] and the proportional reduction in cardiovascular death, myocardial infarction, and stroke resulting from allocation to ramipril, 10 mg daily, was similar in patients with impaired kidney function and those without (HR, 0.79 vs. HR, 0.80; $P > .2$ for heterogeneity). An individual patient data metaanalysis of 26 trials, which included 152,290 participants, has shown that the benefits of reducing blood pressure among patients with eGFR less than 60 mL/min/1.73 m^2 were similar to those among patients with a higher eGFR.[253] However, most participants with CKD had stage 3A, and only 1% of participants had an eGFR less than 30 mL/min/1.73 m^2. An individual patient data metaanalysis has confirmed that the proportional effects of blood pressure lowering in patients with an eGFR less than 60 mL/min/1.73 m^2 are similar to those observed in patients with preserved kidney function.[253]

SPRINT (Systolic Blood Pressure Intervention Trial) compared a systolic blood pressure target of less than 120 mm Hg versus less than 140 mm Hg among 9361 participants at risk of cardiovascular disease, of whom 2646 had CKD at baseline. The trial was terminated early because the primary outcome (acute coronary syndrome, stroke, heart failure, or cardiovascular death) occurred significantly less frequently among those participants assigned the more intensive blood pressure target (5.2% vs. 6.8%; HR, 0.75; 95% CI, 0.64–0.89).[254] The effect was similar among participants with and without CKD.

However, most patients with CKD had relatively mild disease (average eGFR, 47 mL/min/1.73 m^2). A subsequent subgroup analysis of participants with CKD has confirmed that the lower systolic blood pressure target is associated with a reduction in all-cause mortality (HR, 0.72; 95% CI, 0.53–0.99) and cardiovascular events (HR, 0.81; 95% CI, 0.63–1.05), with no overall increase serious adverse events, although there was a greater decrease in GFR after 6 months (–0.47 vs. –0.32 mL/min/1.73 m^2 per year), and AKI, hypokalemia, and hyperkalemia were increased.[255]

Less information is available about the effects of lowering blood pressure on cardiovascular outcomes among patients with more advanced CKD, with most studies involving CKD stages 3b to 5 and primarily assessing kidney endpoints. The Irbesartan Diabetic Nephropathy Trial compared the effects of irbesartan, 300 mg daily, amlodipine, 10 mg daily, and placebo among 1715 subjects with type 2 diabetes mellitus and serum creatinine level of 1.0 to 3.0 mg/dL in women and 1.2 to 3.0 mg/dL in men, principally to assess the effects of each regimen on CKD progression. Irbesartan was shown to be effective at preventing the composite kidney outcome of doubling the serum creatinine level, dialysis, or death.[214] However, neither irbesartan nor amlodipine was superior to placebo in preventing the composite cardiovascular outcome of cardiovascular death, myocardial infarction, congestive heart failure, stroke, or coronary revascularization, despite a 6/3-mm Hg reduction in blood pressure in the irbesartan group and a 4/3-mm Hg reduction in the amlodipine group.[214]

Similarly, the Reduction in Endpoints in non–insulin-dependent diabetes mellitus (NIDDM) with the Angiotensin 2 Antagonist Losartan (RENAAL) study failed to demonstrate a reduction in cardiovascular morbidity and mortality, defined as myocardial infarction, stroke, first hospitalization for unstable angina or heart failure, arterial revascularization, or cardiovascular death.[215] Both these studies had limited power for detecting moderate effects on cardiovascular outcomes, but additional information on the effects of lowering blood pressure among people with advanced CKD is available from metaanalyses of trials conducted exclusively among patients on dialysis, with one including a total of 1202 patients from five trials[256] and the other 1679 patients from eight trials.[257] Both analyses have concluded that blood pressure–lowering therapy reduces cardiovascular event rates and cardiovascular death in patients on dialysis when compared with controls. In the more complete meta analysis, a reduction of 4 to 5 mm Hg in mean systolic and 2 to 3 mm Hg in diastolic blood pressure was associated with a 29% reduction in both cardiovascular events and cardiovascular mortality when compared with control regimens (HR, 0.71; 95% CI, 0.55–0.92).[257] This is comparable to the effect that would be expected from a blood pressure reduction of this magnitude in the general population.

Although a number of randomized controlled trials of blood pressure–lowering interventions have been conducted in kidney transplant recipients, these have almost exclusively examined transplantation-related outcomes such as graft loss, change in glomerular filtration rate, and proteinuria and have not assessed cardiovascular endpoints.[258]

Clinical Relevance

Blood pressure is likely to be a key causal risk factor for cardiovascular disease among patients with CKD. Because patients with CKD typically have higher blood pressure, treatment may well be associated with significant risk reductions. Although uncertainties about precise details of how intensively to treat blood pressure among patients with CKD remain, it is likely that significant gains could be made with treatment to current targets, and further gains may be possible with more intensive treatment.

Thus, based on our present knowledge, it is sensible to treat hypertension in patients with CKD, although there is still much to be learned about the optimal timing of blood pressure measurements, the thresholds for treatment, and the safest method to achieve the largest possible reduction in blood pressure.[259] Older guidelines, including those published by the U,S. National Kidney Foundation, have recommended a target blood pressure of less than 130/80 mm Hg in patients with CKD.[260] However, because there are few studies that have addressed the optimal blood pressure goal in CKD, this target is largely based on data from other high-risk groups, such as patients with diabetes mellitus or congestive heart failure. A systematic review of 11 trials that assessed different targets in patients with advanced CKD has suggested that achieving a blood pressure less than 140/90 mm Hg may be adequate in nonproteinuric patients, with no additional benefits associated with tighter blood pressure control.[261] In observational studies, current targets are achieved in fewer than 50% of CKD patients, possibly in part because more intensive blood pressure control is associated with an increased number of adverse events.[262]

An international guideline published by the Kidney Disease: Improving Global Outcomes (KDIGO) study in 2012 has recommended aiming for a blood pressure less than 140/90 mm Hg in nonproteinuric patients with CKD who are not undergoing dialysis.[263] If proteinuria (UACR >3 mg/mmol) is present, the guideline recommends a lower blood pressure target of less than 130/80 mm Hg, although this is based on posthoc analyses and long-term follow-up of cohorts previously randomized to different blood pressure targets. Data from the general population support more intensive blood pressure lowering and suggest that the potential benefits among patients with CKD are substantial. Therefore the uncertainty surrounding the optimal target should not distract nephrologists from making efforts to reduce blood pressure in patients (especially the very large number of patients with systolic blood pressures well above 140 mm Hg) who frequently are undertreated and are therefore left at unnecessary risk.[262]

DYSLIPIDEMIA

Other than in the presence of nephrotic range proteinuria, the blood LDL cholesterol concentration is not normally raised in patients with CKD. Nevertheless, randomized trials in the general population have clearly shown that reducing LDL cholesterol reduces the risk of myocardial infarction (MI) or death from coronary artery disease, ischemic stroke, and coronary revascularization, even among people with average or low blood cholesterol levels.[129] Therefore even though dyslipidemia is probably not the main contributor to an increased risk of atherosclerosis in CKD as compared with persons without CKD (see earlier, "Dyslipidemia"), reducing the blood LDL cholesterol level may be an effective strategy for reducing such risk, and a number of randomized trials have addressed this hypothesis.

Among persons without established kidney disease, posthoc analyses of randomized trials conducted in the general population have shown that the proportional reduction in the risk of a MI or CAD death, stroke, or coronary revascularization is independent of kidney function.[129] As discussed earlier, however, by the time patients reach stages 4 and 5 CKD, the pathophysiology of cardiovascular disease may have been modified by the "uremic" milieu and, as well as an increased risk of atherosclerotic events, there is an increased risk of cardiac arrhythmia and heart failure related to arteriosclerosis.

Two randomized trials have examined the effects of lowering the LDL cholesterol level with statin therapy on cardiovascular outcomes, specifically among patients receiving dialysis. In the 4D (Die Deutsche Diabetes Dialyse Study) randomized trial of atorvastatin, 20 mg daily, versus placebo among 1255 patients on maintenance hemodialysis with type 2 diabetes, lowering the LDL cholesterol level by an average of about 39 mg/dL (1.0 mmol/L) for a median of 4 years yielded a nonsignificant 8% reduction in the prespecified primary outcome of cardiac death, nonfatal MI, or stroke.[97] In the AURORA (A study to evaluate the Use of Rosuvastatin in subjects On Regular hemodialysis: an Assessment of survival and cardiovascular events) randomized trial of rosuvastatin, 10 mg daily, versus placebo among 2776 patients on maintenance hemodialysis (≈25% of whom had diabetes mellitus), lowering the LDL cholesterol level by an average of about 43 mg/dL (1.1 mmol/L) for a median of 4 years yielded a nonsignificant 4% reduction in the primary outcome of cardiovascular death, nonfatal MI, or stroke.[96] A high proportion of cardiac deaths in these trials were not attributable to CAD, and there was no evidence in either trial of a reduction in the risk of such deaths (Fig. 54.6).[96,97] However, although lowering the LDL cholesterol level with statin therapy among patients with ESKD did not yield statistically significant reductions in the primary outcomes in these trials, there were promising proportional reductions of 18% (risk ratio [RR], 0.82; 95% CI, 0.68–0.99; P = .03) in major cardiac events in the 4D trial and of 16% (RR 0.84; 95% CI, 0.64–1.11; P = .2) in nonfatal MI in the AURORA trial.[96,97] These findings raised the possibility of small, but meaningful, proportional

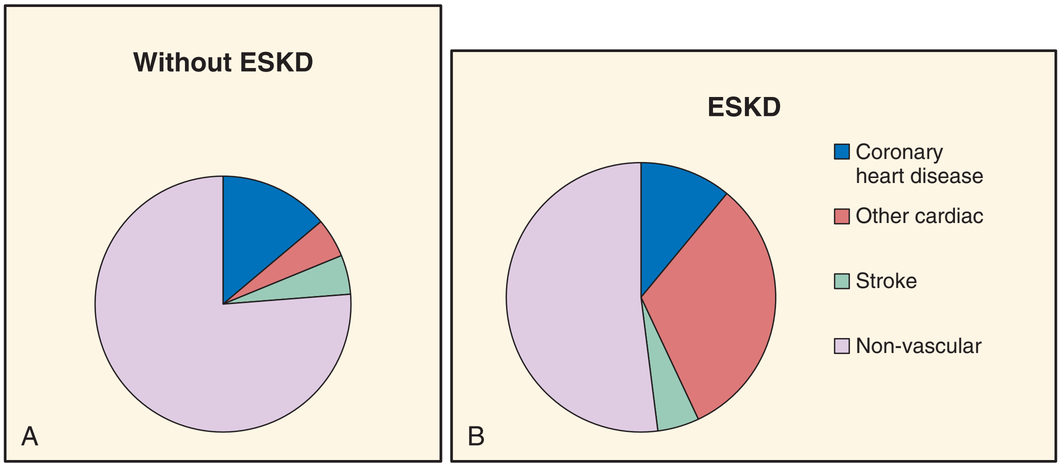

Fig. 54.6 Causes of death in individuals with and without end-stage kidney disease (ESKD). (A) Shown is a group of patients without overt kidney disease recruited into trials of statin therapy (CTT; Cholesterol Treatment Trialists' collaboration) compared with patients with ESKD (B) as recorded by the US Renal Data System (USRDS) or recruited into the Die Deutsche Diabetes Dialyse (4D) trial or into A study to evaluate the Use of Rosuvastatin in subjects On Regular haemodialysis: an Assessment of survival and cardiovascular events (AURORA). Note the declining contribution of coronary artery disease and increasing proportion of other cardiac disease in the ESKD population. (From Baigent C, Keech A, Kearney PM, et al. Efficacy and safety of cholesterol-lowering treatment: prospective meta analysis of data from 90,056 participants in 14 randomised trials of statins. *Lancet* 2005;366:1267-1278; US Renal Data System. USRDS 2009 Annual data report: Atlas of chronic kidney disease and end-stage renal disease in the United States. Bethesda: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2009; Wanner C, Krane V, Marz W, et al. Atorvastatin in patients with type 2 diabetes mellitus undergoing hemodialysis. *N Engl J Med* 2005;353:238-248; Fellstrom BC, Jardine AG, Schmieder RE, et al. Rosuvastatin and cardiovascular events in patients undergoing hemodialysis. *N Engl J Med* 2009;360:1395-1407.)

benefits on atherosclerotic outcomes among patients on dialysis.

Subsequently, the SHARP (Study of Heart and Renal Protection) trial assessed the safety and efficacy of reducing the LDL cholesterol level among patients with stages 3 to 5 CKD, including many who were receiving dialysis. To achieve an average reduction in LDL cholesterol of about 1 mmol/L without using high statin doses (which are associated with an increased risk of myopathy,[264] especially in patients with impaired renal function[265]), a low dose of a statin (simvastatin, 20 mg daily) was combined with a cholesterol absorption inhibitor[9] (ezetimibe, 10 mg daily); the biochemical efficacy and tolerability of this regimen were first confirmed in the UK-HARP pilot studies.[266,267] Overall, 9270 patients were randomized to ezetimibe or simvastatin versus placebo, and allocation to ezetimibe or simvastatin yielded an average LDL cholesterol difference of 0.85 mmol/L (with about two-thirds adherence) during a median follow-up of 4.9 years, and produced a 17% proportional reduction in major atherosclerotic events (526 [11.3%] ezetimibe or simvastatin vs. 619 [13.4%]; RR, 0.83; 95% CI, 0.74–0.94; log rank $P = .0021$; Fig. 54.7).[98]

There was a nonsignificant reduction in nonfatal myocardial infarction or coronary death (213 [4.6%] vs. 230 [5.0%]; RR, 0.92; 95% CI, 0.76–1.11; $P = .37$), and significant reductions in nonhemorrhagic stroke (131 [2.8%] vs. 174 [3.8%]; RR, 0.75; 95% CI, 0.60 – 0.94; $P = .01$) and revascularization procedures (284 [6.1%] vs 352 [7.6%]; RR, 0.79; 95% CI, 0.68–0.93; $P = .0036$). Allocation to ezetimibe or simvastatin was not associated with a significant excess risk of myopathy, hepatitis, or gallstones, and there were no significant excesses of any type of cancer, death from cancer, or death from other noncardiovascular causes.

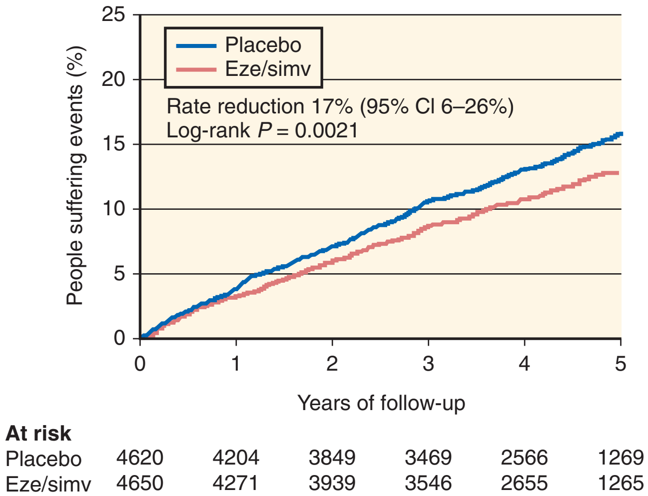

Fig. 54.7 Life table plot of effects of allocation to ezetimibe (eze) or simvastatin (simv) on major atherosclerotic events. *CI,* Confidence interval.

Within SHARP, there was no statistical heterogeneity between those participants on dialysis at baseline and those who were not (P for heterogeneity = .25 for the primary outcome). This finding may at first seem to be discrepant with the earlier studies among dialysis patients. An individual patient data metaanalysis of 28 trials, including over 180,000 participants, showed a significant trend, such that the proportional reduction in major vascular events decreased as renal function declined (see Fig. 54.8).[103] However, it is not possible to determine whether the proportional effect

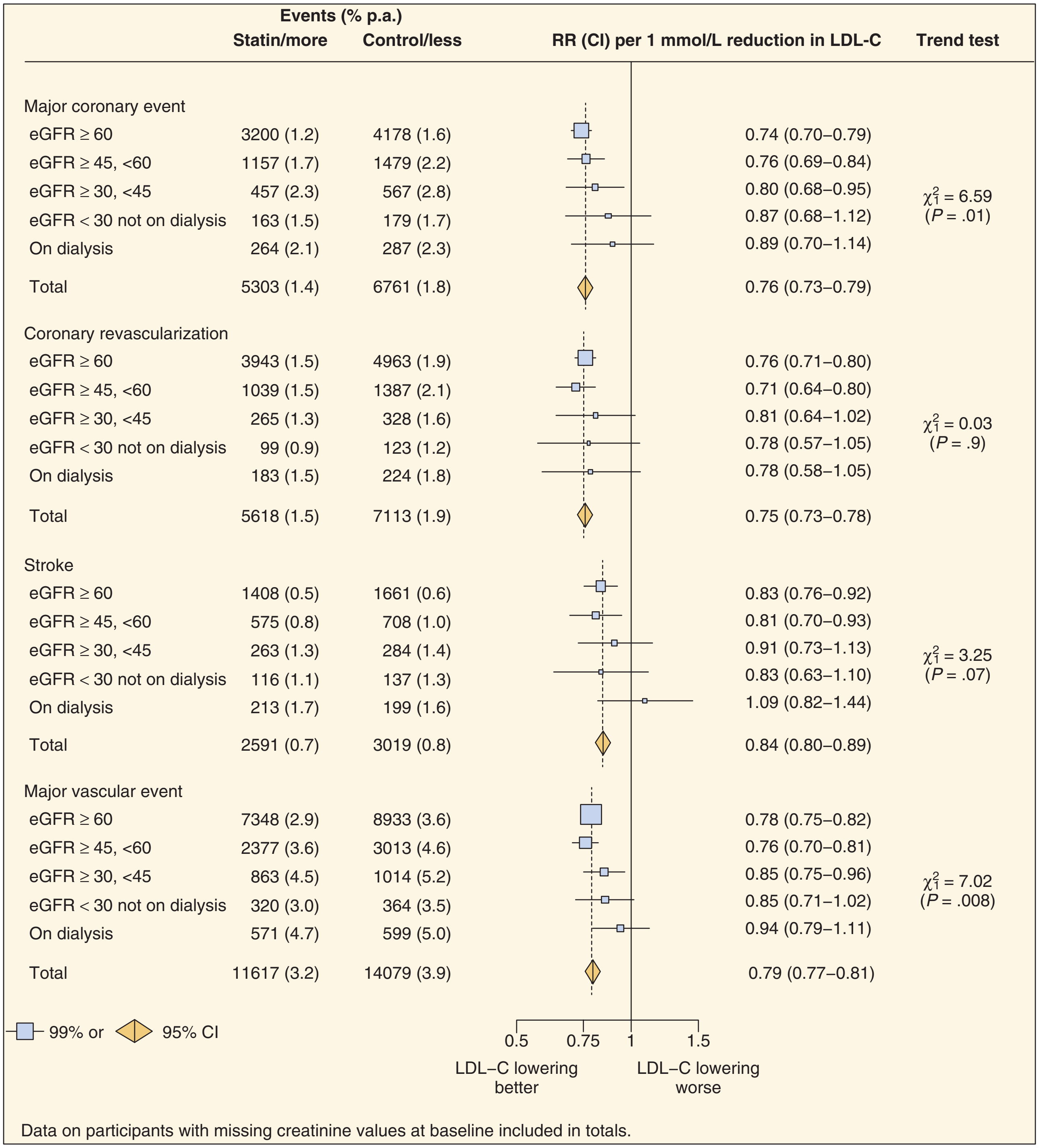

Fig. 54.8 Effects of low-density lipoprotein (LDL)–lowering therapy on particular vascular outcomes in 28 trials by baseline kidney function. *eGFR,* Estimated glomerular filtration rate; *LDL-C,* LDL cholesterol.

gradually diminishes once the eGFR falls below 30 mL/min/1.73 m^2 or whether it is consistent until patients require dialysis, when the proportional effect reduces substantially. A gradual diminution could be explained by the proportion of cardiac deaths that are attributable to coronary disease, which becomes smaller as CKD progresses, and potential misclassification (despite careful clinical outcome adjudication) caused by the atypical presentation of acute coronary syndromes in patients with advanced CKD, especially those on dialysis.

Note that although the proportional benefit may decline as CKD progresses, the absolute risk of atherosclerotic disease increases. Previous metaanalyses have suggested that the risk of a major vascular event increases by 29% for every 30% reduction in eGFR, so a reduction in the eGFR from 60 to 10 mL/min/1.73 m^2 would be associated with a fourfold increased risk of a major vascular event. This is of a similar magnitude to the reduction in the proportional effect of lowering the LDL cholesterol level by 1 mmol/L so that the absolute benefits of doing so would be similar among patients with an eGFR of 10 mL/min/1.73 [2] as those with an eGFR of 60 mL/min/1.73 m.[2] These estimates are based on the results of trials that typically lasted 3 to 5 years, but long-term follow-up of some of these trials has shown that the benefits may accrue beyond this within-trial period.[268,269]

Recent guidelines have emphasized the importance of treating overall risk rather than individual risk factors.[270,271] An international guideline published by KDIGO in 2013 recommended using statins or statin-based lipid lowering regimens in patients with stages 3 to 5 CKD who were older than 50 years and in those younger than 50 years, with additional risk factors.[270] Given the safety and current costs of generic statin-based therapy, the risk threshold at which LDL-lowering therapy should be considered has fallen (to 7.5% over 10 years in recent U.S. guidelines[271]). Although most risk calculators have not been validated in CKD populations, most patients with CKD will have an atherosclerotic risk that warrants treatment, even if their LDL cholesterol level is average or even below average. The potential benefits are substantial, and suggest that widespread use of LDL-lowering therapy in patients with CKD would result in a worthwhile reduction in cardiovascular disease complications.

Clinical Relevance

Reducing the LDL cholesterol level with statin-based therapy should be considered in most patients with CKD. Although the LDL cholesterol level might not be substantially elevated, reducing it will reduce the risk of atherosclerotic disease. To achieve a useful reduction in the LDL cholesterol level, it may be necessary to combine moderate doses of statins with other treatments (e.g., ezetimibe) that lower the LDL cholesterol level.

TIGHT GLYCEMIC CONTROL

Among patients without established CKD, strict glycemic control has been shown in randomized trials to reduce the risk of microvascular disease outcomes (including the development of albuminuria). In type 1 diabetes, a strategy of more intensive glycemic control reduced the development of microalbuminuria by 39% (95% CI, 21%–52%).[211] In type 2 diabetes, the use of a gliclazide-based strategy to reduce hemoglobin A1c (HbA1c) below 6.5% was associated with a reduction in the development of nephropathy (HR, 0.79; 95% CI, 0.66–0.93, $P = .003$).[212] A recent individual patient data metaanalysis has shown that intensive glucose lowering (which reduced HbA1c by 0.9% on average) reduced the risk of nephropathy by 20% (HR, 0.80; 95% CI, 0.72–0.88) and retinopathy by 13% (HR, 0.87; 95% CI, 0.76–1.00).[272] Individual trials in the general population have not shown a clear benefit in terms of cardiovascular risk reduction with stricter glycemic control in type 1 or type 2 diabetes and, indeed, some have even suggested a hazard.[273] However, a metaanalysis of the available randomized trials has supported the hypothesis that stricter glycemic control (with HbA1c lower, on average, by 0.9%) reduces the risk of cardiovascular events, albeit modestly (OR, 0.85; 95% CI, 0.77–0.93).[274]

There is now evidence that certain glucose-lowering mechanisms may be particularly effective at reducing cardiovascular risk. Inhibitors of the sodium-glucose cotransporter 2 (SGLT2i) have been shown to reduce the risk of cardiovascular death among patients with diabetes, with similar effects observed in patients with and without CKD.[275,276] This class of drugs is of particular interest because there is some preliminary evidence to suggest that they may also slow the progression of diabetic kidney disease, which in turn may reduce the risk of cardiovascular disease.[277] Glucagon-like peptide-1 (GLP1) agonists have also been shown to reduce the risk of cardiovascular disease among patients with diabetes,[278,279] so it seems possible that the mechanism of glucose lowering may be important, in addition to the degree of glucose lowering.

It is very likely that the existing trials included a significant proportion of patients with diabetic kidney disease, but the results among such patients have not been reported as a specific subgroup analysis. Therefore the available data suggest that it would be reasonable to aim for tight glucose control among patients with CKD, but with due regard for the potential hazards of too rapid a reduction in HbA1c and hypoglycemia,[273,280] as well as the altered pharmacokinetics of hypoglycemic agents in CKD.[281]

ANEMIA

As noted in the earlier section on anemia, a lower hemoglobin level is associated with an increased risk of structural heart disease. Some nonrandomized studies have suggested that the correction of anemia with recombinant human erythropoietin might lead to improvements in cardiac structure and function,[282,283] which led to the hypothesis that such treatment might improve cardiovascular outcomes among patients with renal anemia. A metaanalysis of nine small trials has suggested that erythropoiesis-stimulating agent therapy might reduce the risk of hospitalization for heart failure, but there were only 114 events among 734 participants in this meta analysis.[284] Four large trials have assessed the impact on mortality and cardiovascular events of partial or full correction of anemia. The first three of these studies investigated whether using two different erythropoietin regimens, normalization of hemoglobin, rather than partial correction, would improve cardiovascular outcomes. Only the more recent TREAT trial (see later) was placebo controlled.

The first study randomized 1233 patients on hemodialysis with clinical features suggestive of ischemic heart disease or heart failure. The trial was stopped early (after a median follow-up of 14 months) predominantly because of an excess risk of vascular access thrombosis with higher hemoglobin concentrations. At this point, 33% of patients had died or experienced a nonfatal MI in the group randomized to a hemoglobin target of 14.0 g/dL as compared with 27% of those randomized to the 10.0-g/dL target (RR, 1.3; 95% CI, 0.9–1.9).[285] The Correction of Haemoglobin and Outcomes in Renal Insufficiency (CHOIR) study, which recruited 1432 patients with CKD not undergoing dialysis and naïve to erythropoiesis-stimulating agent therapy, supported this observation. The trial was terminated because patients randomized to achieve a hemoglobin value of 13.5 g/dL were more likely to reach a primary endpoint—death, myocardial infarction, hospitalization to treat heart failure—than those in the 11.3-g/L target group (HR, 1.34; 95% CI, 1.03–1.74).[286]

The findings of the Cardiovascular Risk Reduction by Early Anemia Treatment with Epoetin Beta (CREATE) trial conducted among patients with stage 3 or 4 CKD were similar; no benefit was demonstrated for targeting a normal hemoglobin (13.0–15.0 g/dL) compared with a subnormal hemoglobin (10.5–11.5 g/dL) and, numerically, more cardiovascular events occurred in the normal hemoglobin group.[287] It has long been recognized that erythropoiesis-stimulating agents (ESAs) can increase blood pressure and blood viscosity (particularly when hemoglobin changes are rapid), and it was predicted over 20 years ago that hemodynamic and rheologic effects may offset the cardiovascular benefits of correcting anemia.[288] Thus in the Trial to Reduce Cardiovascular Events with Aranesp Therapy (TREAT), the first large randomized-controlled trial (RCT) to compare an erythropoiesis-stimulating agent with placebo, the protocol was designed to avoid rapid changes to hemoglobin concentrations.[208] In this study, 4038 patients with type 2 diabetes, CKD, and anemia received either darbepoietin alfa in an effort to achieve a target hemoglobin of 13 g/dL or a placebo with "rescue" therapy if hemoglobin fell below 9 g/dL. The primary endpoint of death or nonfatal cardiovascular event occurred in 31.4% of patients assigned to the active treatment and 29.7% of patients assigned to placebo (HR, 1.05; 95% CI, 0.94–1.17). There was a statistically significant twofold increase in the risk of stroke in the darbepoietin alfa-treated group (101 vs. 53 strokes; HR, 1.92; 95% CI, 1.38–2.68).

Taken together, the results of these four key trials do not support the concept that correction of anemia using ESAs reduces cardiovascular risk in patients with CKD, whether or not on dialysis.[205] However, the current data do not exclude the possibility that small corrections in hemoglobin (e.g., fixed, low-dose, ESAs) are beneficial in the context of CKD. The use of ESAs probably improves health-related quality of life among patients with debilitating symptoms of anemia, although the trials to date are inconclusive on this[205] and, by reducing transfusion requirements, also reduce the risk of sensitizing potential transplant recipients to nonself human leukocyte antigens (discussed in more detail in Chapters 55 and 70).

HOMOCYSTEINE

Randomized trials of folate and B vitamins in both the general population and patients with CKD have shown that reducing the homocysteine level does not reduce the risk of cardiovascular events.[161] The HOST (Homocysteinemia in Kidney and End-Stage Renal Disease) trial randomized 2056 adults with an eGFR less than 30 mL/min/1.73 m^2 (n = 1305) or ESKD (n = 751) to a regimen of folic acid and B group vitamins or placebo.[289] Despite a 25.5% reduction in the plasma homocysteine level, there was no difference in mortality (448 deaths in the treated group vs. 436 in the placebo group; HR, 1.04; 95% CI, 0.91–1.18) nor was there a reduction in cardiovascular events, including MI, stroke, and amputation of the lower extremity, which were all components of the secondary endpoint. In addition, the Folic Acid for Vascular outcomes in Transplantation (FAVORIT) trial randomized 4110 kidney transplant recipients with elevated homocysteine levels and an eGFR of 30 mL/min/1.73 m^2 or higher to a multivitamin tablet containing folic acid, vitamin B_6, and vitamin B_{12} or to a low-dose vitamin B_6 and B_{12} regimen.[290] Despite the expected reduction in homocysteine levels in the treated group, there was no difference in a composite cardiovascular endpoint (adjusted HR, 1.0). A meta analysis of all trials of folic acid in CKD populations has found a similar null result.[163] These results are consistent with the totality of evidence in the general population.[161]

CHRONIC KIDNEY DISEASE AND MINERAL BONE DISORDERS

Although interventions aimed at correcting the biochemical and endocrinologic manifestations of CKD-MBDs were initially driven by the need to prevent the associated debilitating bone pathologies, the suggestion that there might be a link between bone and vascular health in CKD has led to a greater focus on assessing the effects of interventions on cardiovascular outcomes.

Reducing Serum Phosphate

Numerous observational studies have demonstrated a link between increased serum phosphate levels and mortality. However, there have been no placebo-controlled trials investigating the benefits of lowering phosphate levels.[291] It is therefore impossible to know whether current guideline recommendations to maintain serum phosphate concentrations *below* certain targets are beneficial or harmful. Current therapies only reduce the serum phosphate level modestly, although the effects on phosphate balance (as indicated by changes in urinary phosphate excretion) may be larger than suggested by the serum phosphate level alone.[292]

The most widely used phosphate binders are calcium-based, and there are some data to suggest that their use contributes to vascular calcification. Patients with advanced CKD and ESKD taking calcium-based phosphate binders are in positive calcium balance; only patients with ample residual kidney function on loop diuretic agents remain in neutral or possibly negative calcium balance. The availability of non–calcium-based phosphate binders (e.g., sevelamer, lanthanum, newer iron-based compounds) has allowed this hypothesis to be tested, but trials designed to assess whether sevelamer retards the progression of vascular calcification as compared with calcium-based phosphate binders have yielded conflicting results.[172,293,294, 295] The Dialysis Clinical Outcomes Revisited (DCOR) trial compared sevelamer and calcium-based binders among 2103 patients on maintenance hemodialysis.[294] The primary outcome was all-cause mortality and was similar in both arms (HR, 0.93; 95% CI, 0.79–1.10), although nearly half of the trial participants discontinued their assigned treatment before the end of the trial, and over 200 were lost to follow-up. It is therefore unclear whether to treat elevated serum phosphorus concentrations and, if so, with what agent and to what target.

Reducing the serum phosphate level might be expected to reduce FGF23 concentrations. However, a randomized trial comparing sevelamer, lanthanum, and calcium-based phosphate binders found that overall (i.e., data pooling all binders ["active therapy"] vs. placebo), reducing the serum phosphate level by 0.3 mg/dL had no effect on FGF23 concentration.[292] However, whereas FGF23 increased in patients assigned calcium-based binders (median increase, 28 pg/mL; $P = .03$), it decreased in patients assigned sevelamer (median decrease, 24 pg/mL), and was unchanged in patients assigned lanthanum.

Niacin also reduces the serum phosphate level by about 0.4 mg/dL via inhibition of the intestinal sodium-phosphate transporter.[296] However, recent randomized trials have shown that niacin does not reduce cardiovascular risk, despite multiple potentially beneficial effects—lipid modification and blood pressure reduction, in addition to effects on phosphate balance—and, indeed, has a number of harmful effects.[136,297] These trials do not therefore provide support for the use of niacin for reducing the serum phosphate level or for other indications to reduce cardiovascular risk in CKD or ESKD.

Vitamin D

1,25-dihydroxyvitamin D (calcitriol) and its synthetic analogues (collectively termed "active vitamin D derivatives") have been used to correct vitamin D deficiency in the context of CKD for many years. These compounds replenish calcitriol, which is produced by the kidney and suppress levels of PTH, preventing the development of sHPT.[298] More recently, attention has focused on deficiency of parent vitamin D (usually measured indirectly as 25(OH)- vitamin D [25(OH)D], the hydroxylated compound produced in the liver). Over 80% of patients with CKD have vitamin D insufficiency or frank deficiency.[299] In patients not undergoing dialysis, low levels of 25(OH)D are more strongly associated with progression to ESKD and death than calcitriol (1,25-$(OH)_2$D) levels.[300] Two small trials of paricalcitol (an active vitamin D compound) have failed to demonstrate any effect on cardiac structure assessed by cardiac MRI.[301,302] To date, no trials of vitamin D (in any form) have been designed specifically to investigate clinical outcomes such as fractures, cardiovascular events, or mortality among patients with CKD, and a meta analysis of 76 randomized trials of vitamin D has found that insufficient data are available on cardiovascular outcomes to assess any possible reduction in the risk of such events.[303] Despite observational data suggesting that vitamin D supplementation may be beneficial, there is no randomized trial evidence to support recommending such treatment for the reduction of cardiovascular events among patients with CKD.

Calcimimetics

Calcimimetic treatment reduces serum PTH and calcium concentrations.[304] The Evaluation of Cinacalcet Hydrochloride Therapy to Lower Cardiovascular Events (EVOLVE) trial has compared a cinacalcet-based regimen with a non–cinacalcet-based regimen (including a placebo) in 3883 patients undergoing hemodialysis with moderate to severe sHPT.[305] The primary composite outcome was time to death, nonfatal MI, hospitalization for unstable angina, heart failure, or a peripheral vascular event. In the intention to treat analysis, allocation to cinacalcet was associated with a nonsignificant 7% reduction in the primary outcome (HR, 0.93; 95% CI, 0.85–1.02; $P = .11$).[206] However, the randomization process was only stratified for country and diabetes status and, by chance, the two groups were imbalanced for age (median age, 55.0 years, among those allocated cinacalcet vs. 54.0 years in those allocated placebo). After adjustment for age, allocation to cinacalcet was associated with a nominally significant reduction in the primary outcome (HR, 0.88; 95% CI, 0.79–0.97; $P = .008$) and all-cause mortality alone (HR, 0.86; 95% CI, 0.78–0.96; $P = .006$).

The treatment effect was similar for all components of the primary outcome and on vascular and nonvascular mortality. Taking all fatal and nonfatal cardiovascular events combined, randomization to cinacalcet was associated with a 16% lower hazard of nonatherosclerotic cardiovascular events(95% CI, 4%–26%). Although the hazard of atherosclerotic events was also numerically lower in patients randomized to cinacalcet (95% CI, −1%–24%), this difference was not significant. Although derived from posthoc analyses, these data support the hypothesis that mineral bone disease disturbances contribute to nonatherosclerotic cardiovascular disease.

Cinacalcet reduces PTH in patients with CKD not on dialysis.[306] However, the associated reduction in the serum calcium level is more marked than in patients undergoing dialysis (with 62% of those assigned cinacalcet having a serum calcium < 8.4 mg/dL compared with 6% of those assigned placebo). This has led to safety concerns that have limited the provision of cinacalcet to patients on dialysis or, rarely, to patients with other parathyroid disorders (e.g., parathyroid carcinoma, primary hyperparathyroidism (HPT) in a patient unsuitable for surgery).[306]

ANTIPLATELET THERAPY

Antiplatelet therapy reduces the risk of occlusive cardiovascular disease among patients at high risk of such disease, and these benefits greatly outweigh any risks of bleeding,[307] but patients with CKD are at increased risk of bleeding.[308] Thus it is unclear whether the balance of benefit and hazard remains beneficial among such patients. The Anti-Thrombotic Treatment (ATT) trialists' collaborative metaanalysis of trials of antiplatelet therapy among patients at high risk of occlusive vascular disease included 14 trials conducted among 2632 patients on maintenance hemodialysis after placement of a dialysis graft or fistula. Overall, antiplatelet therapy yielded a 41% proportional reduction in the risk of nonfatal MI, nonfatal stroke, or vascular death, which was consistent with the proportional benefit observed among other high-risk patients studied, but there was insufficient data on bleeding risk to assess safety.[307] Based on event rates among patients receiving dialysis in SHARP (and applying the summary risk ratios for aspirin from the ATT metaanalysis),[309] treating 1000 patients on dialysis with vascular disease with aspirin for 5 years has been projected to cause an additional 19 intracranial bleeds and 53 serious extracranial bleeds. There is therefore substantial uncertainty about the relative benefits and harms of antiplatelet therapy in ESKD.

There have been no appropriately sized trials of aspirin with clinical endpoints in patients with CKD not requiring dialysis treatment. Among patients with early-stage CKD, posthoc analysis of the Hypertension Optimal Treatment trial has shown that the benefits of aspirin were similar in the 470 patients with a serum creatinine level higher than 117 μmol/L (1.5 mg/dL) than those with a lower serum creatinine level, but did not report information on any additional bleeding risks.[310] However, there is very little information about the effects of aspirin among those with more severe CKD. One pilot study has demonstrated that aspirin increases the risk of minor bleeding in patients with CKD, but there was no detectable increase in the risk of major bleeding.[266,307] Randomized trials will be required to evaluate the relative risks and benefits of antiplatelet therapy at different stages of CKD.

MINERALOCORTICOID ANTAGONISM

Mineralocorticoid antagonists (MRAs) have been shown to reduce cardiovascular mortality and morbidity among patients with heart failure and reduced ejection fraction.[311–313] It is also likely that MRA treatment improves these outcomes among patients with heart failure and preserved ejection fraction, although the evidence is less clear.[314,315] Given the phenotypic similarity between heart failure and cardiovascular disease in CKD, there is a good rationale to test MRAs among patients with CKD. However, MRAs are known to increase potassium concentrations, so the safety of this among patients with reduced GFR is uncertain. However, patients on dialysis are at high risk of heart failure and at reduced risk of hyperkalemia because of their regular dialysis and lack of functioning nephrons to conserve potassium.[316] The outcomes of large trials of MRAs in dialysis patients are therefore of considerable interest.[317]

DIALYSIS DELIVERY

For patients with CKD who are undergoing hemodialysis, there has been interest in whether modifying the dialysis technique could ameliorate the associated cardiovascular risk. Most hospital-based hemodialysis is delivered on a thrice-weekly schedule, which includes a long gap of 3 days each week. Observational studies have shown that the risk of death is highest at the end of this long gap, most likely due to the accumulation of fluid and toxins during this period.[89] Trials of more frequent dialysis have suggested a benefit on surrogate outcomes such as left ventricular mass, but have not been large enough to determine effects on clinical outcomes.[318] The HEMO study tested high- versus low-flux hemodialysis and more versus less intensive dialysis (i.e., a Kt/V_{urea} target of 1.45 vs. 1.05), but did not demonstrate a benefit of either intervention on clinical outcomes.[319] High-flux dialysis did not demonstrate a benefit in other trials.[320] On-line hemodiafiltration (OL-HDF, which offers very high-flux dialysis) was reported to be associated with a reduced mortality risk in a metaanalysis,[321] but when further trials were included in an updated metaanalysis, this apparent benefit was no longer apparent.[322] The clinical benefit of the improved removal of so-called middle molecules by OL-HDF (compared with standard hemodialysis) therefore remains to be proved.

CONCLUSIONS

Patients with CKD are at increased risk of atherosclerotic and nonatherosclerotic heart disease, as well as stroke and peripheral arterial disease. The increased risk of cardiovascular disease begins early in the natural history of CKD, so an understanding of that excess risk should begin with an assessment of the magnitude of any early changes in known causal risk factors. Hypertension seems likely to be the most important such risk factor because its control is disturbed early in the development of CKD. With CKD progression, other abnormalities, including disorders of mineral metabolism (elevated FGF23, PTH, and altered calcium and phosphate metabolism) appear and may contribute to the rising risk of cardiovascular disease, but there is a lack of evidence about the mechanism whereby these processes might occur.

Observational studies of particular risk factors among patients with advanced CKD cannot be expected to produce reliable information because, as in other sick populations (e.g., very old adults), the observed associations are distorted by confounding and may yield findings that are misleading. For example, there have been numerous observational studies suggesting that cholesterol is inversely associated with major outcomes among patients on dialysis, and yet in the randomized SHARP trial, an unconfounded assessment of the effects of reducing cholesterol has indicated that reducing the LDL cholesterol level is beneficial. For many of the risk factors considered in this chapter, whether established as causal (e.g., hypertension) or of uncertain relevance (e.g., disorders of mineral metabolism), there are no adequately powered randomized trials that assess whether currently available (and often widely used) treatments that modify these abnormalities can safely reduce the risk of cardiovascular events. Studies such as SHARP, TREAT, and EVOLVE have shown that international collaboration can yield really large and informative trials, but many more of comparable size are now needed if effective strategies for reducing the risk of cardiovascular disease are to be identified.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. Which of the following are commonly observed characteristics of cardiovascular disease in patients with chronic kidney disease (select all that apply)?
   a. Arterial stiffness
   b. Left ventricular hypertrophy
   c. Endothelial dysfunction
   d. Valvular heart disease
   e. Dysrhythmias

   Answer: a, b, c, d, e

   Rationale: All of the above phenomena are observed in patients with chronic kidney disease.

2. For which of the following exposures is there good epidemiological evidence of a *causal* association with cardiovascular disease (select all that apply)?
   a. Blood pressure
   b. Low density lipoprotein cholesterol
   c. Serum phosphate
   d. Homocysteine
   e. Anemia

   Answer: a, b

   Rationale: These are the only two risk markers for which there is randomized controlled trial data that reducing the exposure reduces the risk. The other markers only have evidence of association which may be confounded.

3. For which of the following treatments is there good direct evidence of benefit of primary prevention of cardiovascular disease among patients who have received a kidney transplant (select all that apply)?
   a. ACE inhibitors
   b. Aspirin
   c. Statins
   d. Folic acid
   e. Metformin

   Answer: c

   Rationale: Statins are the only treatment that have been shown to reduce risk in a randomized controlled trials in a transplant population.

4. Which of the following exposures are associated with vascular calcification (select all that apply)?
   a. Serum phosphate
   b. Calcium-based phosphate binders
   c. Parathyroid hormone
   d. Fibroblast growth factor-23
   e. Homocysteine

   Answer: a, b, c

   Rationale: Fibroblast growth factor-23 and homocysteine are not associated with vascular calcification.

5. Genetic "Mendelian" randomization studies have supported the inference of a causal nature of the association between which of the following exposures and cardiovascular disease (select all that apply)?
   a. Inflammation
   b. Homocysteine
   c. High density lipoprotein cholesterol
   d. Phosphate
   e. Blood pressure

   Answer: a, e

   Rationale: Mendelian Randomization studies have not found evidence for a causal association between HDL-cholesterol or homocysteine and risk. Mendelian Randomization studies have not been conducted for phosphate.